

Separations on the Job Ladder

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Abstract

We document a U-shaped relationship between within-job wage changes and separations in administrative data from Italy, Germany, and Austria. To interpret this relationship, we estimate a random-search model with worker and firm heterogeneity, learning, persistent match productivity, and wage renegotiation, using Italian data. For Italy, matching the U-shape attributes the decline in wage volatility primarily to learning and implies a 14.5-year half-life for match productivity. Imperfect information lowers output by 0.229 percent through misallocated mobility. Omitting the U-shape moments attributes the volatility decline to negotiation history, shortens the half-life to 3.5 years, and reduces the output loss to 0.073 percent.

Keywords: Separations, Wage Dynamics, Learning, Job Ladder

JEL codes: E24, E25, J24, J31, J63, J64

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1 Introduction

Early careers are marked by frequent job changes and volatile wage growth. Rates of job mobility and the dispersion of wage changes both decline with experience.¹ These parallel profiles are central facts about career dynamics, but they do not reveal why wage growth stabilizes.

Two economic mechanisms can produce this stabilization. Learning makes wage revisions smaller as workers and firms accumulate information about productivity. Wage renegotiation following outside offers can also make wage changes less dispersed as workers accumulate wage claims. The mechanisms differ, however, in how wage revisions are associated with subsequent mobility. Distinguishing them matters because learning changes expected match surplus and the allocation of workers across jobs, whereas accumulated wage claims affect how a given surplus is divided. The same decline in wage volatility can therefore imply different mobility decisions and output losses. This paper uses that distinction to study the sources and allocative consequences of early-career wage instability.

Using administrative employer–employee data from Italy, Germany, and Austria, we document a U-shaped relationship between within-job wage changes and subsequent job separations. Separation is more likely after unusually large wage losses or gains than after moderate wage changes. Relative to the wage-growth decile with the lowest separation rate, the estimated odds of separation are 20–40% higher in the bottom decile and 10–21% higher in the top decile. The left branch accords with standard accounts of deteriorating match quality; the right branch departs from the monotonic negative relationship between wage growth and separation emphasized by [Topel and Ward \(1992\)](#). The U-shape is present in all three countries and is robust to alternative wage measures, controls for career stage and employer size, and separate analyses of job-to-job and employment-to-nonemployment transitions.

The pooled U-shape can, in turn, be decomposed into two statistical components. The between-tenure component reflects composition across career stages: early-career workers experience both more dispersed wage changes and higher separation rates, placing more separations in the tails when career stages are pooled. The within-tenure component captures the relationship among workers at the same career stage: among workers at the same tenure, both large wage losses and large wage gains are associated with higher subsequent separation. Our exact decomposition shows that the within-tenure component accounts for 79.7% of the Italian curvature and also dominates in Germany, whereas between-tenure composition

¹See [Topel and Ward \(1992\)](#) on early-career mobility and [Meghir and Pistaferri \(2004\)](#); [Guvenen \(2009\)](#); [Altonji, Smith Jr. and Vidangos \(2013\)](#) on the lifecycle decline in wage-change dispersion.

dominates in Austria. In Italy, most of the estimated curvature is accounted for by job-to-job transitions rather than transitions to nonemployment.

The decomposition separates familiar lifecycle covariation from a within-career-stage link between wage revisions and subsequent reallocation. The latter is especially informative about mechanisms: revisions in beliefs about worker ability and match productivity can change both wages and the value of the incumbent match relative to outside options, whereas wage claims inherited from earlier outside offers change how a given match surplus is divided between the worker and the firm.

To quantify these mechanisms, we develop a random-search model with on-the-job search, learning about portable worker ability, persistent match productivity, and wage renegotiation following outside offers, building on [Pinheiro and Visschers \(2015\)](#) and [Jarosch \(2023\)](#). Because output reflects both portable ability and persistent match productivity, histories reveal the two components only gradually. Sequential wage renegotiation allows outside offers to leave lasting wage claims. Together, these features generate links between wage revisions and subsequent mobility within tenure, and their joint distribution disciplines the relative importance of information and negotiation history.

The common U-shape coexists with cross-country variation in its composition. We therefore use the three datasets for two distinct purposes: Germany and Austria provide independent replications of the reduced-form relationship, while Italy supplies the quantitative setting. Italy’s uncensored wages and longer panel provide particularly clean measures of the wage-change tails and employment histories central to estimation. All structural estimates and counterfactuals below pertain to Italy.

The preferred estimate, which targets the U-shape together with wage, transition, and selection moments, reproduces the aggregate curvature, its between- and within-tenure decomposition, and the rise in separation in the upper tail. The two branches have different economic content in the estimated economy: large wage losses accompany deterioration in the incumbent match, whereas offer acceptance and job-to-job mobility rise again among workers with the largest wage gains.

To isolate the empirical discipline provided by the U-shape, we reestimate the model after omitting its five moments while retaining the sixteen wage, transition, and selection moments. The reestimation fits those moments better, but it generates an inverted U-shape and selects a different economy. Innovations in match productivity are larger and less persistent, with a half-life of 3.5 rather than 14.5 years, and employed workers receive fewer offers. Under the preferred estimate, greater persistence makes output histories less revealing about fixed

worker ability and keeps current match productivity relevant for future surplus. The U-shape therefore distinguishes an economy with larger, short-lived match-productivity innovations from one in which smaller, persistent innovations have lasting consequences for job value and mobility.

Our main quantitative result is that these changes in the estimated process reverse the economic account of declining wage volatility. Under the preferred estimate, learning contributes 86% of the lifecycle decline in within-job wage-change dispersion among stayers. When the U-shape moments are omitted, accumulated negotiation history instead contributes 94% of the decline. The estimates also assign different meanings to large wage gains: offer acceptance and job-to-job mobility turn upward in the upper tail under the preferred estimate but continue to decline without the U-shape moments. Relative to their respective full-information benchmarks, imperfect information lowers aggregate output by 0.229% under the preferred estimate and by 0.073% without the U-shape moments.

These results matter for quantitative models of idiosyncratic earnings risk. [Guvenen et al. \(2021\)](#) document that earnings changes are non-Gaussian and vary systematically over the lifecycle, while [Hubmer \(2018\)](#) demonstrates that labor-market choices can generate higher moments often interpreted as risk. For models that use estimated earnings processes as inputs, matching wage histories alone can reproduce declining volatility while assigning very different persistence and output costs to the underlying process. Combining wage changes with subsequent employment transitions therefore provides additional discipline on the earnings risk relevant for mobility and aggregate output.

The paper first contributes to the empirical literature on wage growth and separations. Existing work generally finds a negative or weak conditional relationship between wage growth at the current job and subsequent separation.² We document a non-monotonic relationship that remains after accounting for career stage. Our finding is also distinct from the U-shaped occupational-mobility pattern in [Groes, Kircher and Manovskii \(2015\)](#): it concerns changes in a worker’s wage within an ongoing job and subsequent employer separation, and remains after controlling for relative wage position within the worker’s occupation.

The paper also contributes to models of learning and career dynamics. Classic models connect wage revisions and mobility to learning about match quality or worker ability ([Jovanovic, 1979](#); [Gibbons and Waldman, 1999](#); [Prat, 2006](#); [Liu, 2019](#)). [Lange \(2007\)](#) estimates the speed

²See, among others, [Bingley and Westergaard-Nielsen \(2006\)](#), [Galizzi and Lang \(1998\)](#), [McLaughlin \(1991\)](#), [Kim \(1999\)](#), [Kahn and Griesinger \(1989\)](#), [Bartel and Borjas \(1981\)](#), [van der Klaauw and Dias da Silva \(2011\)](#), and [Solnick \(1988\)](#).

of learning, while [Kahn and Lange \(2014\)](#) show that evolving worker productivity can sustain forecast uncertainty even after substantial initial learning. Persistent match productivity plays an analogous moving-target role here: output histories must distinguish fixed worker ability from an evolving match component. [Pastorino \(2015, 2024\)](#) show that learning affects wages indirectly by improving the assignment of workers to jobs within firms. We study the analogous margin across employers: learning changes which incumbent matches survive and which outside offers workers accept.

Finally, the paper relates to equilibrium job-ladder models in which outside offers and employer competition shape wage growth within and between jobs ([Postel-Vinay and Robin, 2002](#); [Cahuc, Postel-Vinay and Robin, 2006](#)). Whereas [Bagger et al. \(2014\)](#) decompose wage growth into human-capital and job-search components, we use the joint behavior of wage revisions and mobility to distinguish learning from wage claims generated by outside offers. The resulting estimates discipline the allocative consequences attributed to imperfect information.

Sections 2–4.2 present the data, empirical relationship, and decomposition; Section 5 develops the model; and Sections 6 and 7 present the estimation and its implications for wage dynamics and allocation.

2 Data and Sample Construction

This section describes the administrative linked employer-employee data used in the empirical analysis and the construction of the harmonized worker-year panel. We combine three datasets from Italy, Germany, and Austria. The three data sources differ in institutional setting, sampling design, period coverage, and wage measurement, so a common empirical pattern across them is informative about the generality of the result. Online Appendix [OA.1.1](#) provides additional details.

2.1 Administrative Data Sources

We use three administrative linked employer-employee datasets: the Italian Veneto Worker History (VWH), the German Linked Employer-Employee Data from the IAB (LIAB), and the Austrian Social Security Database (AMDB). All three datasets are derived from social security records and share a common linked worker-employer structure, but they differ in coverage and design.

The Italian VWH covers employers mainly in the provinces of Treviso and Vicenza in North-East Italy, while the Austrian AMDB covers all of Austria. By contrast, the German LIAB is

based on establishments surveyed in the IAB Establishment Panel and the workers employed in those establishments. Period coverage also differs: the VWH spans 1975–2001, the LIAB 1993–2010, and the AMDB 2000–2016.³

The LIAB contains the richest set of worker and employer observables, including education, occupation, and detailed employer characteristics, although full coworker coverage is available only for 1999–2009. The Austrian AMDB provides comprehensive coworker coverage but more limited worker-level observables. The Italian VWH lies between the two in terms of information content and has the longest time span. Wage reporting also differs across datasets: wages are subject to a reporting ceiling in Germany and Austria, but not in Italy.

2.2 Construction of the Harmonized Worker-Year Panel

We harmonize the three datasets using a common worker-year panel construction similar to [Card, Heining and Kline \(2013\)](#) and [Kline, Saggio and Sølvsten \(2020\)](#).

Main spells and wages. For each worker and calendar year, we identify the highest-paying employment spell and define it as the worker’s main spell. All wages and transitions used in the analysis are defined relative to these main spells. To make wages comparable across datasets, we express earnings in each main spell as real daily wages and work with their logarithm.⁴

Data cleaning. We implement the data-cleaning procedure of [Kline, Saggio and Sølvsten \(2020\)](#): we restrict attention to workers aged 19–63, drop workers with daily wages below 5 real euros, and remove observations with absolute log wage changes greater than one. We also exclude workers with missing sex information, workers ever employed in the public sector, and workers holding ten or more jobs in a given year.⁵ Online Appendix Table [OA.1](#) reports the resulting attrition.⁶

³The employer identifiers are not fully identical across datasets: the LIAB is establishment-based, the VWH is firm-based, and the AMDB uses an administrative employer unit.

⁴For the VWH and AMDB, daily wages are obtained by dividing spell earnings by days worked in the spell. The LIAB directly reports daily wages.

⁵Relative to [Kline, Saggio and Sølvsten \(2020\)](#), we use a less restrictive treatment of missing wage observations. Rather than removing workers entirely when wages are missing in a given year, we exclude only the affected worker-year observation, since wage changes are undefined for that transition. This approach preserves the surrounding employment history and avoids introducing additional sample selection through intermittent non-employment or imperfectly measured unemployment spells.

⁶The sex and entry-age restrictions define the empirical analysis sample but are applied during initial panel construction and therefore appear in this table.

Key variables. Labor-market experience is the number of years in which a worker is observed in a main spell. Tenure is the number of consecutive years at the same employer; the tenure clock resets after a non-employment spell longer than one month, even if the worker later returns to the same employer. Separations are transitions out of the current employer, either to non-employment or to a new employer. Wage growth is $\Delta W_t \equiv W_t - W_{t-1}$, where W_t denotes log daily wages in the worker’s main spell. Employer size is computed from the worker-employer data.

Country-specific observables. The German LIAB records employment type, education, occupation, and detailed employer characteristics, including industry and employer size, and provides administrative tenure and experience measures. In Italy, we observe sector and worker qualification; in Austria, sector but no worker-level characteristics beyond age and sex.

Role in the analysis. The harmonized panel is the starting point for both the reduced-form analysis and the structural estimation. Section 4 uses a restricted empirical analysis sample based on the additional restrictions in Section 2.3, while Section 6.1 returns to the broader Italian worker-year panel to construct the estimation moments. The two samples yield nearly identical separation rates and U-shape estimates (Online Appendix Table OA.20).

2.3 Empirical Analysis Sample Restrictions

We impose additional restrictions to measure within-job wage changes consistently and observe subsequent employer transitions. The restrictions fall into three groups. Online Appendix Tables OA.1 and OA.2 report attrition through harmonized-panel construction and the additional empirical-sample restrictions, respectively.

Worker population. First, because hours worked are not observed in the Italian and Austrian data, an earnings change may reflect either a change in the wage rate or a change in hours. To reduce this concern, we restrict the analysis to men, for whom full-time employment is substantially more prevalent.⁷ This concern is especially relevant early in the career, when labor-force attachment is more variable. In the LIAB, where part-time status is observed, we directly assess the importance of this issue in Online Appendix OA.1.2.3 by excluding part-time spells. We further restrict attention to workers who enter the data before age 30.

⁷In 2010, 28.7% of employed women in the EU worked part-time, compared with 6.5% of employed men. The corresponding country-specific figures were 45.9% and 8.5% in Germany, 28.5% and 5.2% in Italy, and 45.2% and 8.5% in Austria (Eurostat, 2026b).

This allows us to measure tenure and experience from the beginning of the career and to follow workers over a common early-career horizon.⁸ We then exclude apprentices, whose contracts follow distinct training and wage-setting rules. Their earnings dynamics are therefore not directly comparable to those of regular employment relationships.

Wage measurement and timing. To ensure that measured wage changes reflect changes in daily earnings rather than differences in annual exposure, we retain only observations preceded by two periods of full-year employment. We thereby follow continuing worker–firm relationships across consecutive observed records but do not require every record to belong to an uninterrupted sequence of calendar years. When adjacent records can plausibly bracket the same separation, we use days worked to assign the event to the more likely year. This restriction complements our use of daily wages, which divide spell earnings by days worked in Italy and Austria and are reported directly in Germany. We relate separation in period t to the lagged wage change $\Delta W_{t-1} \equiv W_{t-1} - W_{t-2}$, thereby excluding severance payments and other nonstandard earnings components recorded around the separation period. This timing convention follows [Liu \(2019\)](#). We also trim the top and bottom 2 percent of the wage-change distribution.

Observable separations. We exclude observations from a worker’s final year in the data, for which the subsequent transition is unobserved. These observations might be connected to non-standard separations and workers’ exit of the labor force. We also exclude separations associated with employer disappearance or mass layoffs, since these events are more likely to reflect employer-level shocks than worker–employer match dynamics. Following [Schmieder, von Wachter and Bender \(2012\)](#), we define a mass layoff as an event in which an employer with more than 50 employees separates from more than 30% of its prior-year workforce.

2.4 Summary Statistics for the Empirical Analysis Samples

Online Appendix Tables [OA.3](#) and [OA.4](#) report summary statistics for the baseline empirical analysis samples. The three country samples are broadly comparable along the dimensions most relevant for the analysis. Median worker age ranges from 31 in Italy to 33 in Germany. Median tenure is 6 years in all three countries, while median labor-market experience is 9 years. Average annual wage growth is highest in Italy (2.94%) and lowest in Germany (1.95%). Average separation rates are 7.43% in Italy, 8.05% in Germany, and 8.10% in Austria. These

⁸Reassuringly, in the LIAB, our constructed experience and tenure measures are highly correlated with the corresponding administrative measures, with correlations of 0.80 and 0.84, respectively.

levels are broadly consistent with the existing literature.⁹

3 Empirical Strategy

We relate separation from employer j in year t to worker i 's lagged within-job wage change, $\Delta W_{i,j,t-1}$, using the timing and sample defined in Section 2.3.

We document the U-shaped relationship using three complementary specifications. First, we present non-parametric binned estimates that impose minimal structure. Second, we estimate a logit with wage-change-bin indicators, which expresses the pattern in odds-ratio terms relative to the lowest-risk bin. Third, we estimate a quadratic linear probability model (LPM), which summarizes the relationship with a linear term and a curvature term and allows the inclusion of controls.

Non-parametric estimates. We begin by estimating average separation rates within decile bins of the wage-change distribution, without imposing a functional form or adding covariates. Let $k = 1, \dots, 10$ index decile bins of $\Delta W_{i,j,t-1}$. For each bin k , we compute

$$\hat{p}_k = \frac{1}{N_k} \sum_{i,j,t} \mathbf{1}\{\Delta W_{i,j,t-1} \in k\} \text{Separation}_{i,j,t}, \quad (1)$$

where $N_k = \sum_{i,j,t} \mathbf{1}\{\Delta W_{i,j,t-1} \in k\}$. These estimates trace out the conditional separation probability across the wage-change distribution with minimal structure.

Logit odds ratios. Next, we estimate a logit model with indicators for wage-change bins. Let b denote the bin with the lowest average separation rate in the sample, which serves as

⁹Using the EU Labour Force Survey for 2014, [Symeonaki, Stamatopoulou and Karamessini \(2018\)](#) report separation rates of 11.85% for Italy and 12.5% for Austria among workers aged 15–24; the somewhat lower rates in our sample are consistent with its higher average age. For Italy, the mean wage growth of 2.94% is close to the 2% annual growth documented for 1981–1982 in [Galizzi and Lang \(1998\)](#). For Germany, [Lluis \(2005\)](#) reports an average within-job wage growth rate of 1.92% for men with 0–10 years of tenure in the GSOEP for 1985–1996, closely matching our estimate of 1.95% over a more recent period. [Anger \(2011\)](#) similarly finds average wage growth of 2–2.4% in Germany for 1984–2005 among workers aged 20–60. The average employer size in Germany is substantially larger than in Italy and Austria, reflecting the oversampling of large employers, cf. [Fischer et al. \(2014\)](#). Employer sizes in Italy and Austria are very similar. Because of its geographic focus on a highly industrialized region, the Italian dataset has the highest share of manufacturing observations at approximately 56%, followed by Germany at 54% and Austria at 39%. For comparison, [Grinza \(2021\)](#), using a similar vintage of the VWH, reports 65% of her sample in manufacturing.

the omitted category. The specification is

$$\Pr(\text{Separation}_{i,j,t} = 1 \mid \Delta W_{i,j,t-1}) = \Lambda \left(\alpha + \sum_{k \neq b} \mathbf{1}\{\Delta W_{i,j,t-1} \in k\} \beta_k \right), \quad (2)$$

where $\Lambda(\cdot)$ denotes the logistic cdf. For each bin $k \neq b$, e^{β_k} is the odds ratio of separation relative to the reference bin b . We do not include additional covariates in the baseline logit. In nonlinear probability models, adding regressors changes the scale of the latent index, so coefficients are not directly comparable across specifications (Allison, 1999). In the robustness analysis, we instead estimate analogous logits using deciles of residualized wage growth, after partialling out sector, occupation (where available), year, tenure, and experience.

Linear probability model. Finally, we estimate a quadratic linear probability model,

$$\text{Separation}_{i,j,t} = \alpha + \beta_1 \Delta W_{i,j,t-1} + \beta_2 (\Delta W_{i,j,t-1})^2 + \mathbf{X}'_{i,j,t-1} \boldsymbol{\beta}_x + \eta_{i,j,t}, \quad (3)$$

where $\mathbf{X}_{i,j,t-1}$ includes lagged log wages, lagged experience, lagged tenure, lagged employer size, and lagged employer-size growth. The coefficient β_2 captures the curvature of the relationship between wage growth and separation. A positive β_2 indicates convexity. We use the LPM because it provides a parsimonious summary of the relationship and allows coefficients to be compared across specifications with different control sets. Since equation (3) imposes a symmetric quadratic form, β_2 should be interpreted as summarizing the average curvature across positive and negative wage changes.

4 Empirical Results

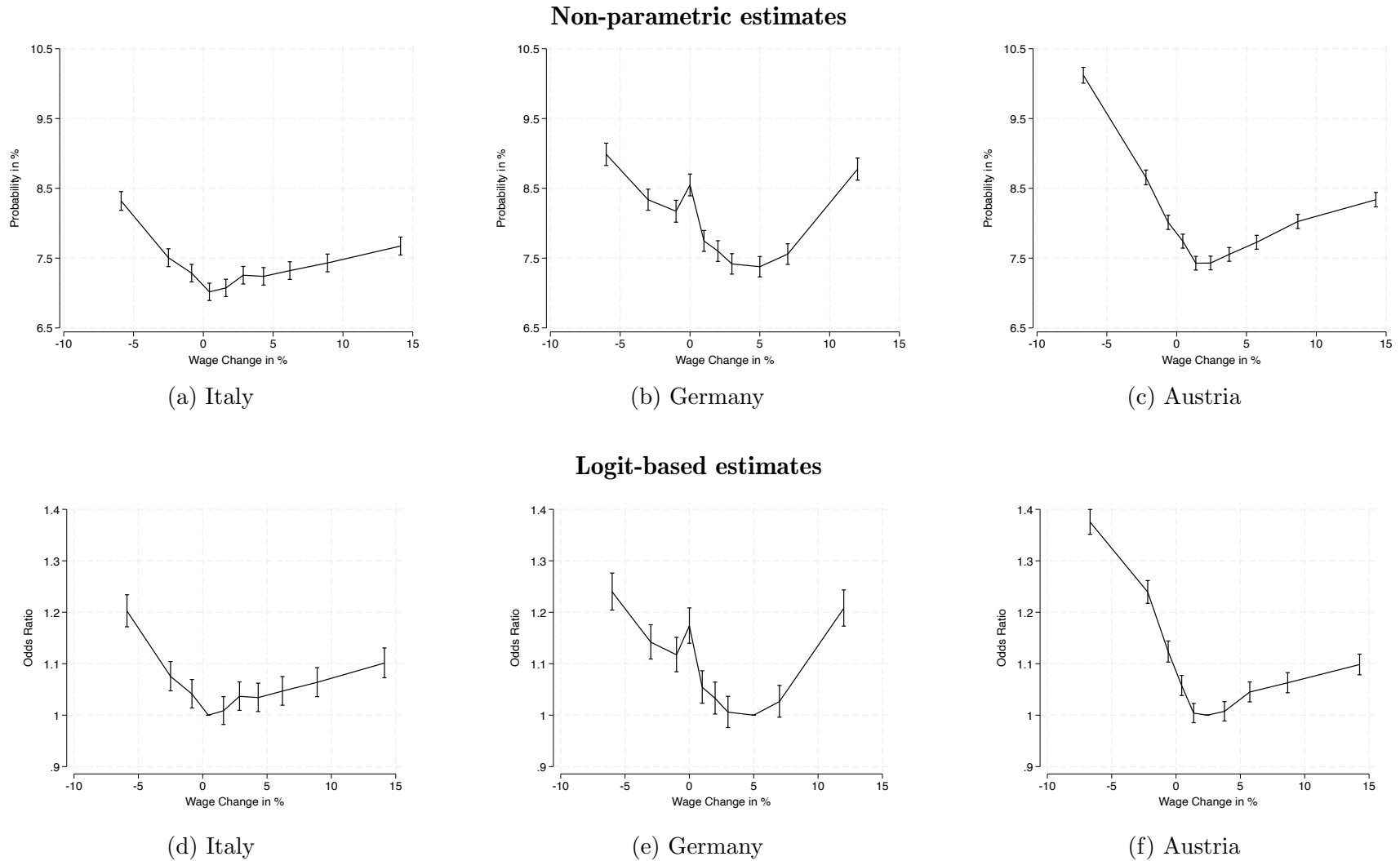
This section documents a U-shaped relationship between within-job wage changes and subsequent separation in Italy, Germany, and Austria. We first establish its shape, magnitude, and robustness across countries in Section 4.1. We then decompose the Italian curvature into variation between and within career stages and summarize the five features carried into structural estimation in Section 4.2.

4.1 Evidence across Countries

Figure 1 plots separation rates by wage-change decile and the corresponding odds ratios relative to the decile with the lowest separation rate. In all three countries, separation is lowest at moderate positive wage growth and rises toward both tails of the distribution.

Relative to the decile with the lowest separation rate, estimated separation odds in the bottom decile are 20 percent higher in Italy, 24 percent higher in Germany, and 40 percent higher in Austria. In the top decile, they are 10, 21, and 13 percent higher, respectively. The right tail is the main departure from [Topel and Ward \(1992\)](#), who document a monotone negative relationship between wage growth and separation in U.S. data. Online Appendix Table [OA.6](#) reports the corresponding logit estimates.

Figure 1: Wage Changes and Subsequent Separations: Non-Parametric and Logit-Based Estimates



Notes: The top row plots mean separation rates in year t by decile of the wage change in year $t - 1$ (Equation (1)). The bottom row plots odds ratios of separation relative to the decile with the lowest separation rate, from the logit in Equation (2); Online Appendix Table OA.6 reports the estimates. Vertical bars show 95% confidence intervals. In the top row, they are based on the standard error of the mean separation rate within each decile; in the bottom row, they are computed on the log-odds scale. In both rows, worker-year observations are treated as independent. The reference decile has no error bar by construction. Wage changes are measured in log points. Each decile contains about 165,000 observations in Italy, 123,000 in Germany, and 276,000 in Austria. Panels are aligned across countries.

Table 1 reports quadratic linear probability models. The baseline specifications control for lagged wages; the augmented specifications additionally condition on experience, tenure, employer size, and employer-size growth. In the baseline specifications, the fitted slope, $\hat{\beta}_1 + 2\hat{\beta}_2\Delta W_{t-1}$, is negative at a 10-log-point wage cut and positive at a 10-log-point gain in all three countries. The estimated troughs, $-\hat{\beta}_1/(2\hat{\beta}_2)$, lie between 2.7 and 6.1 log points, leaving 23 to 46 percent of observations on the rising branch. The additional controls attenuate this rise, particularly in Austria. At a 10-log-point gain, the fitted slope remains positive in Italy and Germany; in Austria, it turns positive only beyond a wage gain of approximately 14 log points. Online Appendix Tables OA.8 and OA.7 and Figure OA.1 report the complete fitted profiles.

Table 1: Wage Changes and Subsequent Separations: Linear Probability Model

	Italy		Germany		Austria	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔW_{t-1}	-0.051*** (0.005)	-0.086*** (0.005)	-0.092*** (0.006)	-0.150*** (0.006)	-0.100*** (0.004)	-0.120*** (0.004)
ΔW_{t-1}^2	0.960*** (0.042)	0.840*** (0.041)	1.290*** (0.063)	1.080*** (0.063)	0.850*** (0.030)	0.420*** (0.030)
W_{t-1}	-0.061*** (0.001)	-0.029*** (0.001)	-0.027*** (0.001)	0.003** (0.001)	-0.066*** (0.001)	-0.040*** (0.001)
Experience $_{t-1}$		0.0004*** (0.0001)		0.0002* (0.0001)		0.0019*** (0.0001)
Tenure $_{t-1}$		-0.0056*** (0.0001)		-0.0073*** (0.0001)		-0.0091*** (0.0001)
Obs.	1,645,579	1,645,579	1,229,924	1,229,924	2,758,316	2,758,316

Notes: The table reports coefficients from linear probability models of separation in year t on lagged wage changes and lagged log wages. The estimating equation is Equation (3). Columns (2), (4), and (6) additionally control for lagged experience, lagged tenure, lagged employer size, and lagged employer-size growth. In Germany, employer size corresponds to observed establishment size. In Italy and Austria, it is constructed as the number of workers attached to the employer identifier in a given year. Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

At a 20-log-point wage gain, the quadratic specification predicts separation rates 1.3 to 3.3 percentage points above those at zero wage growth, whereas the linear specification predicts declines of 0.3 to 1.2 points (Online Appendix Table OA.9). The quadratic effects amount to between one-fifth and more than the full decline in separation between ages 25 and 35 across the three countries (Online Appendix Table OA.5).

The pattern replicates across datasets with different coverage, sampling designs, and employer identifiers. Its presence in the uncensored Italian wage records reduces concern that censoring

in the German and Austrian data generates the right tail. Online Appendix [OA.1.2.3](#) reports eight additional robustness and sensitivity exercises. The estimated U-shape is slightly stronger when we exclude part-time spells in Germany or residualize wage growth with respect to worker and job characteristics. It holds for the subset of the VWH that focuses on fully covered provinces only. It appears separately for employment-to-employment and employment-to-unemployment transitions and, in Germany, for likely layoffs and likely quits; controlling for relative occupational wage position leaves it essentially unchanged. The relationship is more pronounced with contemporaneous wage growth and appears in both East and West Germany. Finally, among job movers, workers in the upper tail of prior wage growth are no more likely to receive a wage increase upon moving, providing little support for a simple search-intensity account of the rising right tail.

4.2 The Anatomy of the Italian U-Shape

The pooled U-shape can reflect two economically distinct relationships. It may be compositional: career stages with more dispersed wage changes may also have higher separation rates, placing more separations in the tails when stages are pooled. Or it may hold within career stages: among workers at similar tenure, large wage losses and gains may both be associated with higher subsequent separation. The distinction matters because the within-stage relationship links individual wage revisions to subsequent mobility after accounting for the lifecycle profiles of wage dispersion and separation. We therefore decompose the Italian curvature into between- and within-career-stage components, computed on the model-comparable sample that also underlies the structural estimation (Section 6.1). We then determine how much of the within component comes from wage losses and whether a rising right-hand branch is supported by economically relevant mass. These facts define the empirical relationship that the model must explain.

Let $X = \Delta W_{i,j,t-1}$ denote the lagged within-job wage change, let Y indicate subsequent separation, let L denote the lagged log wage, and let G index tenure. Let $\hat{\beta}_1$ and $\hat{\beta}_2$ be the coefficients on X and X^2 in the quadratic regression $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_L L + \varepsilon$. Define Q as the residual from projecting X^2 on an intercept, X , and L . The Frisch–Waugh–Lovell theorem and the law of total covariance give the exact decomposition

$$\hat{\beta}_2 = \frac{\widehat{\text{Cov}}(Q, Y)}{\widehat{\text{E}}[Q^2]} = \underbrace{\frac{\widehat{\text{Cov}}(\widehat{\text{E}}[Q | G], \widehat{\text{E}}[Y | G])}{\widehat{\text{E}}[Q^2]}}_{B_G} + \underbrace{\frac{\widehat{\text{E}}[\widehat{\text{Cov}}(Q, Y | G)]}{\widehat{\text{E}}[Q^2]}}_{W_G}. \quad (4)$$

The between component B_G is the covariance between tenure-group means of residualized

squared wage changes and separation. The within component W_G is the observation-weighted average covariance within tenure groups. Equation (4) partitions the estimated quadratic coefficient without changing the regression specification or sample. It is a descriptive covariance decomposition; the model provides its economic interpretation. Online Appendix OA.1.3 gives the derivation and relates the between component to the career profiles of wage-change dispersion and separation.

The between-tenure component accounts for 20.3 percent of the Italian curvature, while the within-tenure component accounts for 79.7 percent. Using experience cells yields a within-cell share of 83.9 percent; using joint tenure-by-experience cells yields 76.8 percent. If differences across career stages were the sole source of curvature, the within component would be zero (Appendix Proposition A.2). The bulk of the Italian U-shape therefore reflects covariation among workers at the same observed career stage and supplies information beyond the marginal career profiles.¹⁰

Complementary regressions yield the same conclusion. Adding a full set of tenure-by-experience indicators retains about 84 percent of the baseline Italian curvature and a positive fitted slope at a 10-log-point wage gain. Allowing the wage-change polynomial to vary across four tenure groups yields positive curvature and a positive slope at that wage gain in every group (Online Appendix Tables OA.10 and OA.11).

To measure asymmetry, we partition W_G over negative, zero, and positive wage changes, using the same residual, tenure-group means, and denominator as in Equation (4). Wage losses account for 70.6 percent of the within-tenure component. We also verify that the fitted quadratic rises on the observed right tail. Its slope at x is $g_x \equiv \hat{\beta}_1 + 2x\hat{\beta}_2$; $g_{0.1}$ is positive, and 12.2 percent of observations have wage changes above 0.10. Table 2 summarizes these dimensions of the Italian U-shape.

¹⁰Online Appendix Table OA.19 reports corresponding decompositions for Germany and Austria. Within-career-stage variation also dominates in Germany, whereas Austria’s curvature arises primarily between career stages. The structural analysis below focuses on Italy.

Table 2: Moments Characterizing the Italian U-Shape

Moment	Data	Bootstrap SE
Overall curvature, $\hat{\beta}_2$	95.62	4.28
Between-tenure share, $B_G/\hat{\beta}_2$	20.31%	0.98
Negative-change share of within curvature, W_G^-/W_G	70.56%	3.29
Slope at $X = 0.10$, $g_{0.1}$	13.98	0.59
Mass above $X = 0.10$, $100 \Pr(X > 0.10)$	12.19%	0.03

Notes: The curvature and slope are measured in percentage points per squared log point and per log point, respectively. Bootstrap standard errors are sample standard deviations across 200 worker-cluster bootstrap draws, in the same units as the targets (Online Appendix Table OA.26). The between-tenure share is $B_G/\hat{\beta}_2$. The negative-change share is W_G^-/W_G , where W_G^- is the contribution of observations with $X < 0$ to the within-tenure component, computed using the same residual, tenure-group means, and denominator as Equation (4). The sample is the Italian model-comparable sample used to construct the structural moments. Equivalent moments for Germany and Austria can be found in Online Appendix Table OA.18.

We use these five statistics as estimation targets because each captures a distinct feature of the empirical relationship: overall curvature, the part that remains within career stages, its concentration among wage losses, and the slope and empirical support of the right-hand branch. Together with the wage, transition, and selection moments, they discipline the model’s account of wage revisions and subsequent mobility. Section 5 develops that account.

5 Theoretical Framework

We develop a random-search model with on-the-job search, learning about portable worker ability, persistent match-specific productivity, and wage renegotiation following outside offers. The exposition proceeds in two steps. Section 5.1 presents a learning benchmark that shows how information accumulation reduces wage revisions and clarifies a lifecycle-composition channel for the pooled U-shape. Section 5.2 presents the model used in estimation, in which persistent match productivity and sequential-auction wage setting allow wage changes and subsequent mobility to covary within tenure. Appendix A.3 presents the filtering equations, and Appendix A.4 gives the complete recursive formulation.

5.1 A Learning Benchmark

Time is discrete. A continuum of risk-neutral firms and workers discount the future at $\tilde{\beta} \in (0, 1)$.¹¹ Workers, indexed by i , have fixed unobserved productivity $a_i \sim V(a)$; firms, indexed by j , have productivity $\mu_j \sim F(\mu)$. Log output of worker i at firm j and time t is

$$y_{i,j,t} = \mu_j + a_i + \eta_{i,j,t}, \quad \eta_{i,j,t} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\eta^2).$$

Workers and firms update beliefs about a_i after observing output. At labor market entry, beliefs are normal with mean $A_0 = E[a_i]$ and variance $\sigma_{a,0}^2$. Posterior beliefs with mean $A_{i,t}$ and variance $\sigma_{a,i,t}^2$ evolve by Bayes' rule:

$$\sigma_{a,i,t}^2 = \frac{\sigma_{a,i,t-1}^2}{1+s_{i,t-1}}, \quad A_{i,t} = A_{i,t-1} + \frac{s_{i,t-1}}{1+s_{i,t-1}} \xi_{i,j,t}$$

where $s_{i,t-1} = \sigma_{a,i,t-1}^2 / \sigma_\eta^2$ is the signal-to-noise ratio, and $\xi_{i,j,t} = y_{i,j,t} - (A_{i,t-1} + \mu_j)$ is the surprise in observed output. As usual in Bayesian updating, belief variance declines deterministically with experience (see Appendix A.1 for the formal argument).

Unemployed workers receive benefits z and job offers with probability λ_0 ; employed workers receive outside offers with probability λ_1 , with offers in both states drawn from $F(\mu)$. Matches dissolve exogenously with probability δ , and workers exit the labor force exogenously with probability Ψ . Write $P_{a,i,t} \equiv \sigma_{a,i,t}^2$ for the posterior variance of ability. We denote firm productivity by $J = \mu_j$ and the worker's pre-output information state by $I_t = (A_{i,t-1}, P_{a,i,t-1})$. In the scalar benchmark, posterior variance is a deterministic function of the number of accumulated output signals.

Within period t , the wage w_t is set at the start of the period from the pre-update state. Output is then realized and beliefs update. An exogenous separation occurs with probability δ ; if the match survives, an outside offer may arrive, after which the worker chooses among unemployment, staying, and switching.

To isolate the learning mechanism, the benchmark uses a spot-wage normalization under transferable utility: each period the worker is paid expected output, so the firm's expected flow profit is zero. We relax this assumption in Section 5.2.

¹¹For expositional convenience, the model uses the term *firm*. In the empirical sections, we use *employer* because the administrative identifier is not fully identical across datasets. Nothing in the model depends on that distinction.

As a result, wages $w(J, I)$ are

$$w(J, I) = \mathbb{E}[y_{i,j,t} \mid J, I] = \mu_j + A_{i,t-1}.$$

In the following, we suppress the subscripts i, j and t when context permits.

Match surplus $S(J, I)$ consists of the net output gain over unemployment, an option value from learning, a continuation value from ongoing matches, and the option value of on-the-job search. Let $U(I)$ denote the value of non-employment and write $x^+ \equiv \max\{x, 0\}$. Formally:

$$\begin{aligned} S(J, I) = & w(J, I) - z + \beta \int (U(I') - U(I)) d\mathcal{G}^{I,J}(I') + \beta(1 - \delta) \int S^+(J, I') d\mathcal{G}^{I,J}(I') \\ & + \beta(1 - \delta)\lambda_1 \int \int (S^+(J', I') - S^+(J, I'))^+ dF(J') d\mathcal{G}^{I,J}(I') \\ & - \beta\lambda_0 \int S^+(J', I) dF(J') \end{aligned} \quad (5)$$

where $S^+(J, I) = \max\{S(J, I), 0\}$, $\beta = \tilde{\beta}(1 - \Psi)$, and $\mathcal{G}^{I,J}$ is the transition kernel for the updated information state I' induced by output at firm J . The model features four types of job separations: exogenous separations to unemployment, endogenous separations to unemployment, job-to-job mobility from viable matches, and job-to-job mobility that occurs because the incumbent match becomes non-viable after learning.

Unemployed workers enter employment if surplus is positive. The unemployment value function is hence:

$$U(I) = z + \beta \left(U(I) + \lambda_0 \int S^+(J, I) dF(J) \right) \quad (6)$$

Given the firm distribution F and the information-state transition kernel $\mathcal{G}^{I,J}$, the model can be solved numerically and used to simulate transitions and wage dynamics.

The benchmark clarifies one source of pooled curvature. Learning reduces the variance of wage revisions as signals accumulate (Appendix Proposition A.1), while job-ladder progression tends to lower separation rates with tenure. If these gradients hold in the selected wage-change sample and squared wage changes are orthogonal to subsequent separation within tenure, pooling tenure groups yields a positive quadratic coefficient due to learning. Appendix Proposition A.2 states these sufficient conditions formally.

5.2 The Estimated Model

The model taken to the data extends the learning benchmark in two directions. First, output contains a persistent match component, so each signal is informative about both portable ability and current match quality (Section 5.2.1). Second, wages follow a sequential-auction contract, so outside offers can change wages while the worker remains with the incumbent employer (Section 5.2.2). Output and offer histories therefore shape wage revisions and mobility jointly. Appendix A.4 gives the complete recursive formulation, including unemployment, joint surplus, value sharing, and wages.

Both learning and wage renegotiation can make wage-change dispersion decline over the career. Learning reduces revisions as beliefs become more precise; outside offers change the worker’s negotiation benchmark and the pass-through of match productivity into wages. Their different implications for subsequent mobility give the joint wage-change and separation moments leverage in disciplining their relative importance.

5.2.1 Information and Match Productivity

In the model used in estimation, log output is

$$y_{i,j,t} = \mu_j + a_i + m_{i,j,t}, \quad m_{i,j,t} = \rho m_{i,j,t-1} + \eta_{i,j,t}, \quad \eta_{i,j,t} \sim N(0, \sigma_\eta^2).$$

A new match begins at the known state $m_{i,j,0} = 0$. The first post-hire observation therefore contains $m_{i,j,1} = \eta_{i,j,1}$, whose prior variance is σ_η^2 . Match productivity is thus an experience good in the sense of Jovanovic (1979): it is revealed through output produced within the match. Beliefs about portable ability carry across jobs, while a job change resets the match state and its covariance with ability. Appendix A.3 gives the joint filtering equations. When a match begins, let $P_{a,ij}^0$ denote the posterior variance of ability carried into that match and define $q_{ij} \equiv \sqrt{P_{a,ij}^0}$. Let $\hat{m}_{i,j,t-1}$ denote the posterior mean of match productivity carried into period t , and let $\tau_{i,j,t}$ denote current tenure. We define

$$J \equiv \mu_j, \quad I_t \equiv (A_{i,t-1}, q_{ij}),$$

and the match state as

$$M_t \equiv (\hat{m}_{i,j,t-1}, \tau_{i,j,t}).$$

For given structural parameters, (J, I_t, M_t) summarizes firm productivity and the beliefs relevant for expected output and joint surplus.¹² We suppress worker, firm, and time subscripts below.

The persistence parameter ρ governs how output histories distinguish portable ability from match productivity. When $\rho = 0$, match productivity is i.i.d.; every output observation provides a fresh noisy signal of ability, and the filter reduces to the scalar learning problem above. When $\rho > 0$, the same output innovation may reflect either ability or persistent match quality, and subsequent observations separate the two components only as match productivity mean-reverts. A job change resets match productivity to zero and therefore provides a fresh signal about the same worker. Persistence consequently slows learning within a job and makes current match quality more important for future surplus. Mean reversion also creates reversals in expected match productivity, which the wage contract maps into serial dependence in wage changes. Appendix A.3 gives the corresponding filter.

5.2.2 Wage Determination

Wages follow the sequential-auction protocol of Cahuc, Postel-Vinay and Robin (2006), implemented recursively as in Jarosch (2023). Let $S(J, I, M)$ denote joint match surplus and let R summarize the worker's bargaining position inherited from hiring and subsequent outside offers. With bargaining weight α , the worker's value above non-employment is

$$b(S, R) \equiv \max\{\min\{R + \alpha(S - R), S\}, 0\}.$$

Joint surplus is invariant to R under transferable utility. For a viable match, normalized negotiation capital $\omega \equiv \min\{R/S, 1\}$ measures the fraction of surplus represented by the inherited benchmark. The contract state R completes the state vector, so (J, I, M, R) is sufficient for wages and continuation values. Appendix A.4 gives the complete recursion.

An outside offer that creates more joint surplus than the incumbent match induces a move; an offer that improves the worker's bargaining position without exceeding incumbent surplus raises R while the worker stays. Negotiation history therefore changes wages while leaving the ranking of available matches unchanged. Learning instead changes expected joint surplus and can affect both wages and mobility. The polar cases make the distinction transparent: at $\alpha = 1$, the worker captures the full match surplus; at $\alpha = 0$, the worker's claim is pinned to R , subject to feasibility. Intermediate values of α determine how strongly changes in match

¹²For given structural parameters, the Kalman covariance recursion yields deterministic functions $P_a(q, \tau)$, $P_m(q, \tau)$, and $C(q, \tau)$. Thus (q, τ) contains the same covariance information as (P_a, P_m, C) . Appendix A.3 gives the recursion.

surplus pass through to wages.¹³

The model allows the career decline in wage volatility to reflect learning, innovations in match productivity, and wage renegotiation. We next estimate their quantitative importance using standard wage and transition moments together with the joint relationship between lagged within-job wage changes and subsequent separation.

6 Estimation for Italy

In the following, we define the model parameters and describe the moments used in estimation (Section 6.1). We then present the estimates and model fit (Section 6.2) and discuss model mechanisms (Section 6.3).

6.1 Parameterization, Moments, and Identification

Data, sample, and timing. We estimate the model on the Italian administrative data described in Section 2. We use Italy for the structural estimation because the VWH provides uncensored daily wages and linked worker–employer histories over a long panel. The estimation uses a model-comparable sample drawn from the same harmonized worker-year panel as the empirical analysis in Section 3. One model period corresponds to one year. The Italian records cover twenty calendar years, from 1982 through 2001, yielding nineteen year-to-year transitions.¹⁴ We therefore set the simulation horizon to $T = 19$ and impose the same horizon in the data by retaining employed-history counts from 1 through 19. Calendar gaps inside a worker’s retained panel span count as non-employment; observations outside that span do not enter the annual non-employment stock.¹⁵

Parameter Set and Normalization. The model is parametrized by the parameter vector Θ

$$\Theta = \{\sigma_\mu, \bar{\sigma}_\eta, \bar{a}, \sigma_a, h, \rho, \lambda_0, \lambda_1, \delta, z, \alpha, \beta\}, \quad h \equiv \sigma_{a,0}/\sigma_a.$$

¹³When $\rho = 0$ and $\alpha = 1$, the estimated model shares the benchmark’s information process and joint-surplus allocation. The spot-wage normalization and the sequential-auction contract can differ in the timing of transfers.

¹⁴Coverage in the earliest years of the VWH (1975–1981) is incomplete; the estimation sample therefore begins in 1982.

¹⁵Note that our administrative earnings records do not allow us to distinguish unemployment from non-participation. Accordingly, throughout the estimation we interpret “unemployment” as non-employment, i.e. the absence of an observed job/earnings record in year t . The hazard E2U therefore measures transitions from employment to non-employment, and the stock u is the fraction non-employed within retained worker spans. We construct the simulated counterparts using the same observability rule in the model.

We jointly estimate the first eleven elements of Θ and calibrate β separately. Because h and σ_a are estimated, initial uncertainty $\sigma_{a,0} = h\sigma_a$ is a derived quantity. Firm productivity is drawn from $F(\mu) = \mathcal{N}(0, \sigma_\mu^2)$, where we normalize $E[\mu] = 0$ because only relative productivity levels are identified. Worker ability a_i is drawn from $V(a) = \mathcal{N}(\bar{a}, \sigma_a^2)$ and initial beliefs satisfy $A_{i,0} \sim \mathcal{N}(\bar{a}, \sigma_a^2 - \sigma_{a,0}^2)$ with $a_i \mid A_{i,0} \sim \mathcal{N}(A_{i,0}, \sigma_{a,0}^2)$. Thus $\sigma_{a,0}$ measures unresolved uncertainty about ability, and h measures this uncertainty relative to cross-sectional ability dispersion. Match productivity m_{ijt} follows $m_{ijt} = \rho m_{ij,t-1} + \eta_{ijt}$ with $\eta_{ijt} \sim \mathcal{N}(0, \bar{\sigma}_\eta^2)$.¹⁶ Search frictions are governed by the annual offer arrival probabilities for nonemployed workers λ_0 and for employed workers λ_1 . Matches end exogenously with probability δ , nonemployed workers receive home-production flow z , and wages are determined by the worker bargaining weight α . We fix the effective annual discount factor at $\beta = 1/1.10 \simeq 0.91$.¹⁷

Moments. We estimate the eleven free elements of Θ by simulated method of moments, matching a vector of labor-market, wage-level, and wage-dynamics moments, together with moments of the wage-change U-shape and selection along the firm ladder. In the data and the simulations, let $E_{it} \in \{0, 1\}$ denote employment, w_{it} log wages, and $\Delta w_{it} \equiv w_{it} - w_{i,t-1}$. Experience x_{it} counts observed years of employment; we use bins $x \leq 3$, $3 < x \leq 6$, and $x > 6$. Table 3 collects the twenty-one targeted moments. Eleven describe unemployment and transition hazards (u , E2U, E2U+E2E), wage levels ($E[w_{t-1}]$, $E[w_t \mid \text{U2E}]$), cross-sectional wage dispersion by experience ($\text{SD}(w_{t-1} \mid x_{t-1} > 6)$, $\text{SD}(w_{t-1} \mid x_{t-1} \leq 3)$), wage-change dispersion for stayers and movers ($\text{SD}(\Delta w_{t-1} \mid \text{stayer}, x_{t-1} \leq 3)$, $\text{SD}(\Delta w_{t-1} \mid \text{stayer}, 3 < x_{t-1} \leq 6)$, $\text{SD}(\Delta w_{t-1} \mid \text{mover}, x_{t-1} \leq 3)$), and serial dependence of wage changes $\text{Corr}(\Delta w_t, \Delta w_{t-1} \mid \text{stayer})$. The remaining ten moments comprise the five U-shape moments introduced in Section 4.2—overall curvature, the between-tenure share, the negative-change share of within-tenure curvature, the slope at a log wage change of 0.10, and the mass above 0.10—and five selection moments. The five selection moments are the firm-rank change gap between movers with wage losses and gains, the origin-rank gradient in residual wage-growth variance, the experience gradient in absolute wage growth, the share of origins in the top firm quintile, and wage–firm-rank mismatch dispersion. Online Appendix Table OA.21 gives the exact samples, timing conventions, and reporting units. Online Appendix OA.2.1 describes the estimator and numerical procedure.

¹⁶In the baseline specification, $\bar{\sigma}_\eta = \sigma_\eta$ is common across firms. We retain the bar to align the notation with the firm-dependent specification in Online Appendix OA.3.

¹⁷To interpret this calibration, we account for permanent labor-market exit using a constant annual exit probability $\Psi = 1/34.75 \simeq 2.88\%$, where 34.75 is the average expected duration of working life for Italian men in 2000–2001 (Eurostat, 2026a). Under the relation $\beta = \tilde{\beta}(1 - \Psi) = (1 - \Psi)/(1 + r)$, these choices imply an annual pure discount rate of $r = (1 - \Psi)/\beta - 1 \simeq 6.83\%$.

Identification. The twenty-one moments discipline the eleven estimated parameters jointly. Because each primitive affects wage setting, mobility, and sorting, no moment block identifies a parameter in isolation. Stocks and transitions are especially informative about offer arrival, exogenous separation, and the outside option; wage levels and dispersion about worker and firm heterogeneity and bargaining; and wage-change dynamics about initial uncertainty, match-innovation volatility, and persistence. The selection and U-shape moments discipline how these forces vary across firms, careers, and the wage-change distribution. Section 6.3 evaluates this joint mapping using local derivatives and restricted reestimations.

6.2 Estimation Results

Overview. The model provides a close fit across the wage distribution, career wage dynamics, worker–firm selection, and the wage-change U-shape. Table 3 shows that average and entry wages are each matched within about 0.05 log points. The model also reproduces wage dispersion among young workers (21.06 versus 22.63 in the data) and experienced workers (28.11 versus 30.84), as well as the career decline in stayer wage-growth dispersion: the model decline is 2.14 log percentage points, compared with 2.60 in the data. Among the selection moments, the wage-variance slope by origin rank is 5.86 versus 6.43, and the origin top-quintile mass is 30.39 versus 30.01.

The model closely matches the overall U-shape curvature and its decomposition. It generates curvature of 96.56, compared with 95.62 in the data. Differences across tenure groups account for 21.09 percent of model curvature and 20.31 percent in the data, while wage losses account for 68.18 and 70.56 percent of the respective within-tenure components. The fitted profile also turns upward in the upper tail. At a log wage change of 0.10, its slope is 5.08 versus 13.98 in the data, and 7.74 percent of model observations lie above this point versus 12.19 percent in the data. Thus, the model reproduces the U-shaped form and its decomposition, with a minimum farther to the right (0.0737 versus 0.0269) and a more gradual upper-tail rise.

The remaining loss is concentrated in a small number of moments. The wage-growth autocorrelation, the right-hand U-shape slope, and young-mover wage-growth dispersion together account for approximately 72 percent of the loss. The model produces less negative wage-growth autocorrelation (-1.58 versus -17.21) and lower young-mover dispersion (10.26 versus 18.28). It also generates a higher employment-to-non-employment rate (5.91 versus 1.86 percent), a higher non-employment rate (8.16 versus 5.99 percent), and a lower overall separation rate (10.99 versus 14.23 percent). The main limitation of the fit is therefore the dynamics surrounding mobility and non-employment, rather than the wage distribution, the

career decline in stayer volatility, or the central U-shape moments.

Table 3: Estimation Moments: Empirical Counterparts and Model Fit

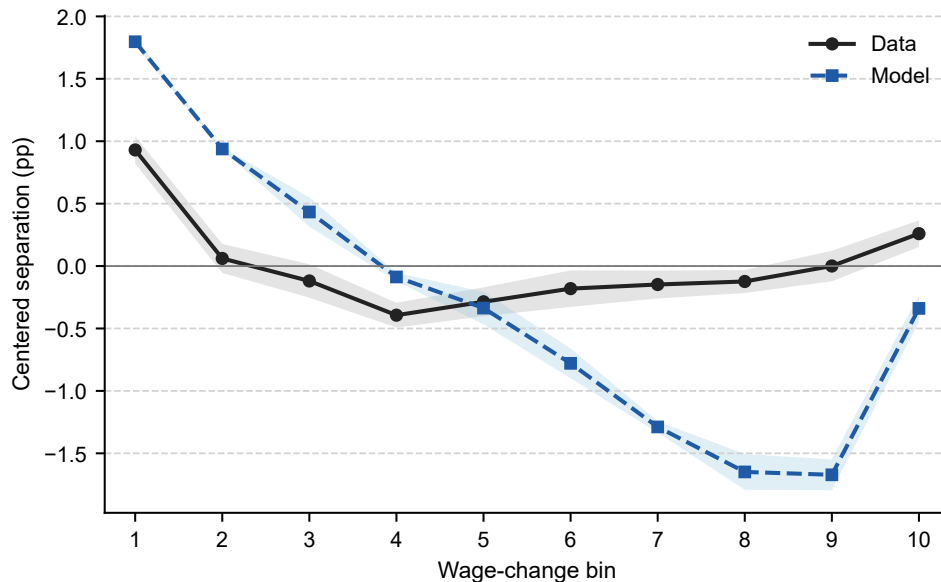
	Moment	Data	Model
U-shape moments	Curvature, β_2	95.62	96.56
	Between-tenure share, $100B/\beta_2$	20.31	21.09
	Negative within share, $100W^-/W$	70.56	68.18
	Slope at +0.10	13.98	5.08
	Mass above +0.10 (percent)	12.19	7.74
Labor market and wage dynamics	Employment to non-employment	1.86	5.91
	Annual non-employment	5.99	8.16
	Overall separation	14.23	10.99
	Mean lagged log wage	480.52	485.62
	SD log wage, high experience	30.84	28.11
	SD wage growth, young stayers	12.11	10.28
	SD wage growth, young movers	18.28	10.26
	SD log wage, young workers	22.63	21.06
	Wage-growth autocorrelation	-17.21	-1.58
	Mean entry log wage from non-employment	469.57	464.16
	SD wage growth, mid-career stayers	9.51	8.14
Selection moments	Firm-rank change gap	-2.55	-4.25
	Wage-variance slope by origin rank	6.43	5.86
	Absolute wage-growth experience slope	-0.94	-0.56
	Origin top-quintile mass	30.01	30.39
	Wage-firm-rank mismatch dispersion	7.49	11.85

Notes: The blocks contain five U-shape, eleven conventional, and five selection/mismatch moments. Wage levels and log-wage dispersion measures are reported in log points multiplied by 100, probabilities and shares in percent, and wage-growth autocorrelation as the correlation multiplied by 100. Online Appendix Table OA.21 reports the remaining units and exact definitions. Model entries average three common-seed moment estimates.

Untargeted Fit. Figure 2 provides a bin-level assessment of the separation profile without imposing a quadratic functional form. In both the data and the model, mean-centered separation declines and then rises in the upper tail, although the model places the trough

farther to the right and generates a larger peak-to-trough movement. A decomposition by observed destination provides a second untargeted check. The E2E curvature component is 81.64 in the model and 78.17 in the data, while the E2U components are 14.91 and 17.44, respectively. Thus, the model reproduces the contribution of both transition types to aggregate curvature.¹⁸

Figure 2: Mean-Centered Ten-Bin Separation Profile



Notes: Bins use fixed empirical wage-change decile cutoffs. Each profile is centered by its own bin-mass-weighted separation rate, so the figure compares shape rather than levels. Model rates average the same three production seeds used for the reported fit. Shaded bands report 95 percent worker-clustered jackknife intervals for the data and 95 percent Monte Carlo intervals across the three production seeds for the model; the latter condition on the estimated parameters.

The transition-specific decomposition also locates the remaining discrepancy. The E2E slope at 0.10 is 3.97 in the model and 11.72 in the data, whereas the corresponding E2U slopes are 1.11 and 2.26. In the U-shape sample, the E2E rate is close (5.81 versus 6.53 percent), while the model E2U rate is substantially higher (5.41 versus 0.98 percent). As Online Appendix Figure OA.6 shows, the muted right-hand rise is concentrated in E2E transitions, while the level overstatement is concentrated in E2U transitions.

¹⁸We reestimate the same specification using the E2E and E2U indicators in turn as outcomes. Because they partition observed separations, their quadratic coefficients sum to aggregate curvature. Classification is by observed destination, so an advance-notice layoff followed by immediate reemployment enters E2E.

Table 4: Parameter Estimates and Local Standard Errors

Parameter	Estimate	Local SE
Firm productivity SD, σ_μ	0.1327	0.00283
Match innovation SD, $\bar{\sigma}_\eta$	0.1225	0.00163
Mean ability, $\bar{a}$	4.9139	0.00623
Ability SD, σ_a	0.2014	0.00385
Initial uncertainty ratio, h	0.5090	0.01440
Nonemployed offer probability, λ_0	0.6640	0.00887
Employed offer probability, λ_1	0.1364	0.00187
Exogenous separation probability, δ	0.0541	0.00053
Bargaining weight, α	0.4665	0.01103
Match productivity persistence, ρ	0.9534	0.00476
Home-production flow, z	4.0341	0.02394
Initial uncertainty SD, $\sigma_{a,0}$	0.1025	0.00413

Notes: Local standard errors combine worker-cluster sampling uncertainty and simulation uncertainty. Online Appendix [OA.2.8](#) describes their construction.

We now discuss the economics of the estimated parameter vector, see Table 4 above, and relate our findings to the existing literature.

Worker and Firm Heterogeneity. Permanent worker heterogeneity exceeds firm heterogeneity: the standard deviation of worker ability, $\sigma_a = 0.201$, is about 51% larger than that of firm productivity, $\sigma_\mu = 0.133$. This ordering is consistent with the earnings decomposition of [Bonhomme, Lamadon and Manresa \(2019\)](#), which assigns a relatively small role to firm heterogeneity in Swedish matched worker–firm data. Mean ability $\bar{a} = 4.914$ anchors the level of output and wages.

Search Frictions and the Job Ladder. The estimated offer arrival probability from non-employment is $\lambda_0 = 0.664$ per year and the employed arrival probability is $\lambda_1 = 0.136$, so employed workers receive offers at about 21% of the nonemployed rate. This ratio is comparable in magnitude to [Lise, Meghir and Robin \(2016\)](#), who estimate relative employed search intensities of 0.332 for workers with high school or less and 0.157 for college graduates on U.S. NLSY data. Our ratio of 0.205 lies between these two estimates. For comparison, [Cahuc, Postel-Vinay and Robin \(2006\)](#) estimate annual employed-offer intensities ranging

from 0.053 to 0.321 across French industries and skill groups. The exogenous layoff probability $\delta = 0.054$ is close to the Italian annual job-loss intensity of 0.0526 reported by [Jolivet, Postel-Vinay and Robin \(2006, Table 2\)](#), equivalent to a 5.1% annual probability at a constant rate.

Bargaining Power. The estimated worker bargaining weight $\alpha = 0.466$ is sizable and implies that workers capture a substantial but incomplete share of the match surplus at hiring. This stands in contrast to the seminal findings of [Cahuc, Postel-Vinay and Robin \(2006\)](#), who detect small bargaining power for intermediate- and low-skilled French workers. It also exceeds the bargaining weights of 0.188 and 0.272 estimated by [Lise, Meghir and Robin \(2016\)](#) for their two education groups. [Jarosch \(2023\)](#) finds a substantially higher value of $\alpha = 0.928$ on German data.¹⁹ Our intermediate estimate of $\alpha = 0.466$ reflects a different set of forces. In our framework, $\alpha < 1$ allows accumulated outside options to affect the pass-through of match-level productivity shocks into wages. The decline in the standard deviation of wage changes over the career—from $\text{SD}(\Delta w \mid \text{stayer}, x \leq 3) = 12.11$ to $\text{SD}(\Delta w \mid \text{stayer}, 3 < x \leq 6) = 9.51$ in the data—disciplines α jointly with learning and the persistence of match productivity. The negative serial correlation in wage changes $\Gamma_1 = \text{Corr}(\Delta w_t, \Delta w_{t-1} \mid \text{stayer})$ provides additional discipline because α and ρ both affect wage dynamics through renegotiation and persistent match shocks.

Outside Option. The home-production flow $z = 4.034$ affects the value of non-employment and hence the reservation rule, accepted matches, and entry wages. Relative to the output of a mean-ability worker at a mean-productivity firm with zero match component, this corresponds to $\exp(z - \bar{a}) \simeq 0.42$. For scale, [Jarosch \(2023\)](#) calibrates the German non-employment opportunity cost to 75% of output.

The Speed of Learning. Initial uncertainty about worker ability is sizable: $\sigma_{a,0} = 0.1025$, or approximately 51% of cross-sectional ability dispersion, and its variance is about 70% of annual match-innovation variance. To compare the implied learning path with the learning literature, we follow [Lange \(2007\)](#) and measure the fraction of the initial assessment error eliminated after three years. At the estimated parameters, the error falls by 41.3% for a worker who remains in the first job. If the worker instead makes one job-to-job move after the first year, it falls by 58.4%, because the new match begins with its match component known to be zero. These two model paths bracket the 50% decline estimated by [Lange \(2007\)](#);

¹⁹In [Jarosch \(2023\)](#), the high α is related to $\lambda_1 > \lambda_0$: low bargaining power would imply low hiring wages as workers compensate employers for the option value of more efficient search in employment, inconsistent with the compressed wage structure in his data.

the stayer path is also close to the roughly one-third decline estimated by [Kahn and Lange \(2014\)](#). The model therefore generates an empirically familiar amount of early-career learning while making its speed depend on mobility history.²⁰

This comparison also clarifies the connection with [Kahn and Lange \(2014\)](#). Their direct performance data imply that firms learn substantially about initial productivity but continue to forecast a moving target because worker productivity itself evolves. In our model, worker ability is fixed, but output reflects both ability and an evolving match component. Its estimated persistence, $\rho = 0.9534$, gives a half-life of approximately 14.5 years, so additional observations within a match separate ability from match quality only gradually, while job changes provide fresh signals. Both frameworks therefore reconcile substantial early learning with persistent forecast uncertainty later in the career. Our model ties that continuing uncertainty to mobility, wage renegotiation, and the U-shaped relationship between wage changes and separations.

6.3 Identification, Mechanisms, and Extensions

We assess identification and mechanisms through three complementary exercises. Local derivatives and a reestimation that excludes the U-shape moments show how those moments discipline the estimate. Restricted reestimations show whether the remaining parameters can compensate for removing a mechanism, while fixed-parameter counterfactuals isolate that mechanism’s role at the preferred estimate. Finally, an extended specification tests firm-dependent signal precision and worker–firm complementarity.

Identification and the Contribution of the U-Shape Moments. The local analysis yields two results. First, the complete moment vector locally identifies all eleven parameters: its moment–parameter Jacobian has full column rank at the estimate. Second, the U-shape moments substantially sharpen identification of parameters governing wage setting and wage dynamics. The signs of the local derivatives accord with the model’s economic channels: mean ability ($\bar{a}$) raises average and entry wages, initial uncertainty ($\sigma_{a,0} = h\sigma_a$) raises wage-growth dispersion among young stayers, and match-innovation volatility ($\bar{\sigma}_\eta$) flattens the right-hand branch of the U-shape. After allowing the remaining parameters to adjust locally, omitting the five U-shape moments makes the profiled loss surface 94.9 percent flatter in the mean-ability direction, 90.6 percent flatter for ability dispersion, 84.2 percent flatter for bargaining power,

²⁰In the linear Gaussian filter, the fraction of an initial assessment error remaining after t signals is $P_{a,t}/P_{a,0}$, where $P_{a,t}$ is the posterior variance of a_i . Using $\sigma_{a,0} = 0.1025$, $\bar{\sigma}_\eta = 0.1225$, $\rho = 0.9534$, and $m_0 = 0$, the estimated filter gives $P_{a,3}/P_{a,0} = 0.587$ in the first-job stayer path and 0.416 following one match reset after the first signal.

76.2 percent flatter for match-innovation volatility, and 75.9 percent flatter for persistence. Online Appendix [OA.2.9](#) reports the complete analysis.

We next reestimate the eleven-parameter model using only the sixteen moments without the U-shape and evaluate the omitted U-shape moments at the resulting estimate. Loss over the sixteen targeted moments falls by 42 percent. Yet evaluating both estimates on the complete 21-moment vector reverses the comparison: total loss is about 197 times as large when the U-shape moments are omitted, driven by a roughly 1,001-fold increase in loss on those five moments. Online Appendix Table [OA.25](#) reports the decomposition. Consistent with this deterioration, the estimated relationship becomes inverted U-shaped, has a negative slope at a log wage change of 0.10, and declines across all ten observed wage-change bins. This reversal concerns a nontrivial part of the distribution: 12.19 percent of wage changes in the data exceed 0.10.

The no-U-shape estimates also describe a different stochastic process. Match-innovation volatility ($\bar{\sigma}_\eta$) rises from 0.123 to 0.290, while persistence (ρ) falls from 0.953 to 0.822, shortening the half-life of a match shock from 14.5 to 3.5 years. Initial uncertainty and permanent ability dispersion also decline, while the employed offer probability falls from $\lambda_1 = 0.136$ to 0.109. Thus, omitting the U-shape moments selects an economy with less initial uncertainty, larger but shorter-lived match fluctuations, and fewer opportunities for employed workers to move. Online Appendix Table [OA.22](#) reports the complete parameter comparison and local standard errors.

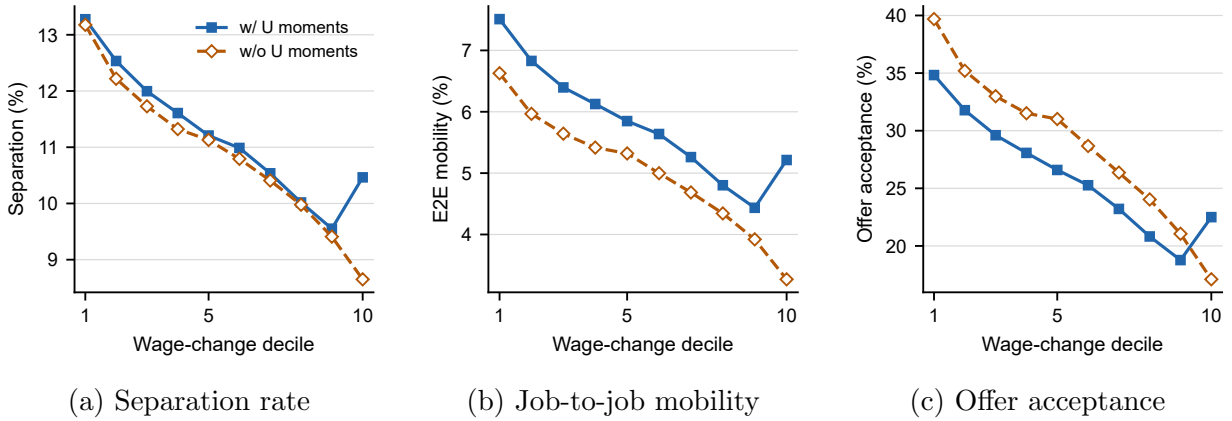
To assess whether this contrast depends on the relative weight assigned to the U-shape moments, we reestimate the model after scaling their joint contribution to loss by one-quarter, one-half, and two. Across these reestimations, match-productivity persistence remains between 0.942 and 0.953, while innovation volatility remains between 0.112 and 0.136, implying half-lives between 11.6 and 14.5 years. Thus, over the tested range, the persistent match-productivity process is robust to the precise positive weight assigned to the U-shape moments; the economically distinct estimate emerges when this information is omitted altogether. Online Appendix Table [OA.23](#) reports the complete comparison.

Online Appendix Figures [OA.5](#) and [OA.7](#) provide untargeted comparisons. The no-U-shape estimate generates a declining raw separation profile despite substantial upper-tail mass. Across tenure, both estimates concentrate too much of the separation decline immediately after tenure year 2; the no-U-shape estimate more closely follows the gradual empirical decline in wage-change dispersion.

Moreover, the U-shape moments change what large wage gains signify. Figure [3](#) shows that,

under the preferred estimate, separation, job-to-job mobility, and on-the-job offer acceptance all turn upward in the top wage-growth decile. From the ninth to the tenth decile, these outcomes rise from 9.55 to 10.46 percent, from 4.43 to 5.21 percent, and from 18.77 to 22.50 percent, respectively. Without the U-shape moments, all three continue to fall. Thus, absent direct discipline from the U-shape moments, large wage gains remain a retention signal. Targeting the U-shape instead selects an economy in which workers in the upper tail return to the mobility margin. Accepted moves lead to higher-ranked firms under both estimates, so the central distinction is whether the largest wage gains coincide with renewed reallocation, not whether those moves upgrade job quality.

Figure 3: Reallocation across the Wage-Change Distribution



Notes: The figure uses ten equal-mass bins of lagged wage changes in each simulated economy’s structural-regression sample. Blue squares report the preferred estimate (w/ U moments); hollow orange diamonds report the reestimation that omits them (w/o U moments). Panels (a)–(c) report the subsequent separation rate, job-to-job mobility rate, and acceptance rate among on-the-job offers in the associated outcome window. Statistics average three simulations of 500,000 workers.

Mechanism Restrictions. Table 5 compares three restrictions under two complementary exercises. The reestimations allow the remaining primitives to adjust and ask whether the economy can compensate for the restricted mechanism. The fixed-parameter exercises impose the same restrictions at the preferred estimate and isolate each mechanism’s direct contribution. Online Appendix Table OA.25 reports fit by moment block.

With the remaining parameters reestimated, near-observed ability, implemented as $\sigma_{a,0} = 10^{-4}$ or $h \simeq 0$, is the best-fitting restricted specification, yet raises loss by approximately 90 percent. It retains an overall U-shape but eliminates the positive between-tenure contribution and weakens the right-hand rise. Setting $\rho = 0$ raises loss by more than 7,000 percent and produces

an inverted U-shape, while setting $\alpha = 1$ raises loss by more than 19,000 percent and leaves an overall U-shape whose right-hand branch declines.

Holding the remaining primitives fixed yields the same qualitative conclusions. Near-observed ability again removes the positive between-tenure contribution and reduces the slope at 0.10 from 5.08 to 2.68; $\rho = 0$ again produces an inverted U-shape; and $\alpha = 1$ reduces the slope at 0.10 to -2.43 . Together, the two exercises support distinct roles for the mechanisms: initial uncertainty shapes the composition and right-hand rise, persistence generates the overall U-shaped form, and wage renegotiation sustains its rising right-hand branch.

Table 5: U-Shape Outcomes under Mechanism Restrictions

Specification	β_2	$100B/\beta_2$	$100W^-/W$	$g_{0.1}$	Right mass
Data	95.62	20.31	70.56	13.98	12.19
Preferred estimate	96.56	21.09	68.18	5.08	7.74
<i>A. Remaining primitives reestimated</i>					
Near-observed ability	95.44	-0.50	73.81	3.09	8.02
$\rho = 0$	-44.37	-2.72	-59.56	4.17	1.76
$\alpha = 1$	16.58	42.00	377.79	-4.30	28.20
<i>B. Remaining primitives held fixed</i>					
Near-observed ability	118.12	-2.12	61.07	2.68	6.38
$\rho = 0$	-62.56	38.95	-86.61	3.55	2.05
$\alpha = 1$	21.54	79.16	917.99	-2.43	19.96

Notes: The data row reports empirical moments; model entries average three production simulations. Panel A reestimates the remaining primitives using all 21 moments. Panel B holds them at the preferred estimate. Near-observed ability fixes $\sigma_{a,0} = 10^{-4}$. A negative β_2 denotes an inverted U-shape, $g_{0.1}$ is the fitted slope at $\Delta w = 0.10$, and right mass is $100 \Pr(\Delta w > 0.10)$. The ratio $100B/\beta_2$ is the between-tenure share of curvature, and $100W^-/W$ is the contribution of wage losses to within-tenure curvature.

Firm-Dependent Learning and Worker-Firm Complementarity. We use an extended specification to test two channels emphasized in related work: firm-dependent signal precision and worker-firm complementarity. The first allows match-innovation volatility, and hence the precision of output signals about worker ability, to vary with firm productivity, as motivated by evidence that learning differs across occupations and with the skills used on the job (Mansour, 2012; Light and McGee, 2015). The second allows the productivity of worker ability to vary across firms, as in models of comparative advantage and worker-firm sorting (Groes, Kircher and Manovskii, 2015; Lise, Meghir and Robin, 2016).

The parameter γ governs firm-dependent match-innovation volatility, while κ governs the loading of worker ability on firm productivity. Both extensions are quantitatively small when all thirteen parameters are reestimated jointly on the same twenty-one moments. The estimates $\hat{\gamma} = 0.0030$ and $\hat{\kappa} = 0.0021$ imply that, between firms at the tenth and ninetieth percentiles of productivity, match-innovation volatility differs by only 0.78 percent and the loading on worker ability by only 0.54 percent. At the extension estimate, imposing $\gamma = \kappa = 0$ raises total loss by just 0.26 percent and slightly improves fit of the five U-shape moments. The additional channels therefore provide no economically meaningful improvement in fit. We retain the additive baseline in which signal precision does not vary with firm productivity. The next section evaluates the implications of learning and persistent match productivity within this parsimonious specification. Online Appendix [OA.3](#) reports the complete comparison.²¹

7 Applications to the Italian Economy

This section traces how targeting the U-shape changes the model’s economic implications in the estimated Italian economy. Section [7.1](#) decomposes the career decline in stayer wage-change dispersion into learning and accumulated negotiation history. Section [7.2](#) then compares each estimated economy with its full-information benchmark to quantify the consequences for mobility and output.

7.1 Why Wage-Change Volatility Declines over the Career

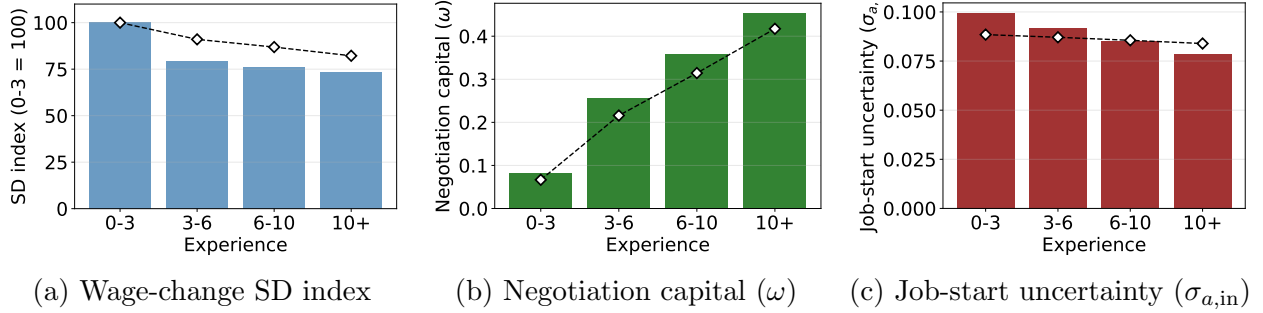
Determinants of wage variability. Learning models connect early-career wage revisions to the arrival of information about worker productivity ([Farber and Gibbons, 1996](#); [Lange, 2007](#); [Kahn and Lange, 2014](#)). Equilibrium job-ladder models instead emphasize how outside offers and employer competition shape wage growth within and between jobs ([Postel-Vinay and Robin, 2002](#); [Cahuc, Postel-Vinay and Robin, 2006](#); [Bagger et al., 2014](#)). Our model contains both channels and allows their quantitative roles to be compared. We summarize these channels by the negotiation benchmark ω_t and the standard deviation of beliefs about ability at the start of the job, $\sigma_{a,\text{in}}$.

Negotiation capital. Let R_t denote the worker’s current contract benchmark and S_t the joint continuation surplus of the incumbent match. For viable matches, define normalized negotiation capital as $\omega_t \equiv \min\{R_t/S_t, 1\}$. A higher ω_t indicates a larger inherited claim relative to current joint surplus and hence a greater role for past outside offers in current

²¹The matched comparison holds the remaining parameters fixed at the estimated extension and measures the joint contribution of γ and κ .

wages. Entrants from non-employment begin with $\omega_t = 0$, while an accepted outside offer can establish a positive benchmark for a job-to-job mover.

Figure 4: Lifecycle profiles of wage-change dispersion and its state variables



Notes: The figure averages three simulated panels of 500,000 workers and 19 yearly periods. Filled bars report the preferred estimate, which targets the U-shape moments; hollow diamonds report the reestimation that omits them. Panel (a) indexes each estimate's cross-sectional standard deviation of stayer wage changes, $SD(\Delta w | \text{stayer}, x)$, to 100 in the 0–3 experience bin. This normalization compares the career decline rather than the different initial levels. Panel (b) reports mean negotiation capital $\omega_t = \min\{R_t/S_t, 1\}$ among the same stayer observations (defined only for viable matches with $S_t > 0$). Panel (c) reports mean job-start uncertainty $\sigma_{a,in}$, the standard deviation of the prior belief about ability at the start of the worker's current job (Appendix A.3).

Job-start uncertainty. With persistent match productivity, a job change resets the prior over the match component and its covariance with ability. Workers with the same experience can therefore begin jobs with different prior uncertainty about ability, depending on their labor-market histories. We summarize this state by $\sigma_{a,in} \equiv q$, the pre-signal standard deviation of beliefs about ability carried into the worker's current job; Appendix A.3 gives the covariance recursion.

Figure 4 reports experience profiles for (a) the indexed career decline in the standard deviation of stayer wage changes, (b) mean negotiation capital ω_t , and (c) mean job-start uncertainty $\sigma_{a,in}$. Under the preferred estimate, stayer wage-change dispersion falls by 26.5 percent over the career, negotiation capital rises from 0.082 to 0.454, and job-start uncertainty falls by 21.0 percent. The no-U reestimation still produces a 17.8 percent decline in wage-change dispersion and a similar rise in negotiation capital, from 0.067 to 0.417, even though job-start uncertainty falls by only 5.1 percent. These comovements show why the lifecycle profile alone cannot identify its mechanism: both channels move over the career, and the same declining wage volatility survives under markedly different information processes. We therefore use a structural decomposition next to understand their relative importance.

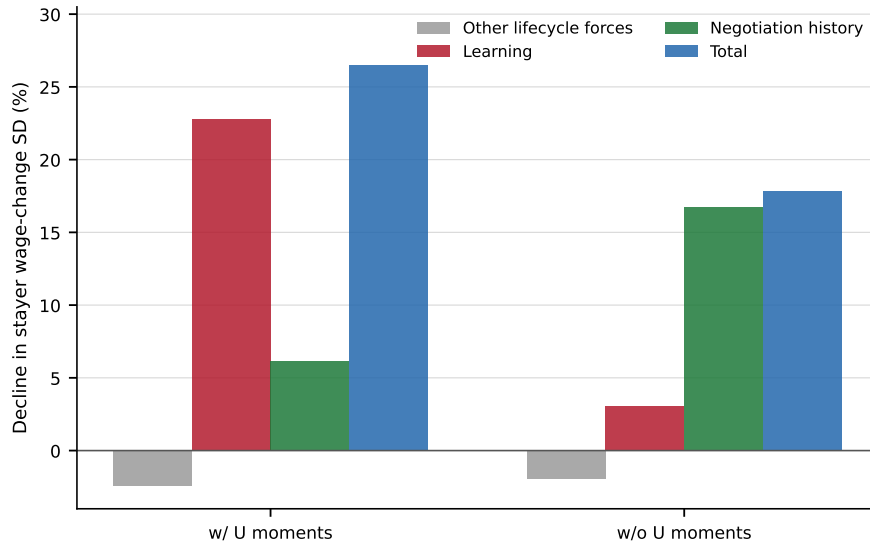
Decomposing wage variability. To quantify the contributions of learning and accumulated negotiation history, we construct a 2×2 counterfactual for each estimate. Along the information dimension, we compare the estimated learning economy with an economy in which worker ability is known at labor-market entry. Along the contract dimension, we recompute the wage at $R_t = 0$ on the same state and allocation histories. Because R_t divides joint match surplus but does not enter the joint-surplus comparisons governing entry, job-to-job moves, or separation, this counterfactual removes the wage effect of accumulated negotiation history without changing allocation.

Let $D_{\ell r}$ denote the percentage decline in stayer wage-change dispersion from 0–3 to more than 10 years of experience, where $\ell \in \{0, 1\}$ indexes known ability versus learning and $r \in \{0, 1\}$ indexes wages computed without versus with accumulated negotiation capital. We report the two-factor Shapley contributions

$$\phi_L = \frac{1}{2}[(D_{10} - D_{00}) + (D_{11} - D_{01})], \quad \phi_R = \frac{1}{2}[(D_{01} - D_{00}) + (D_{11} - D_{10})].$$

By construction, $D_{11} = D_{00} + \phi_L + \phi_R$. The Shapley terms average each channel’s effect over the two possible orders in which the channels are introduced.

Figure 5: Structural decomposition of declining wage-change volatility



Notes: The figure decomposes the percentage decline in the standard deviation of stayer wage changes between the 0–3 and 10+ experience groups. “Learning” and “negotiation history” are the two-factor Shapley contributions defined in the text. “Other lifecycle forces” is D_{00} , the residual lifecycle change generated by selection, mobility, and match-state evolution when ability is known at entry and wages are recomputed at $R_t = 0$. “Total” is D_{11} in the estimated economy. Each object averages three simulations of 500,000 workers under common worker-period shocks.

Figure 5 reveals a sharp reversal in the dominant mechanism. Under the preferred estimate, learning contributes 22.8 percentage points (95% CI: [20.4, 25.1]), or about 86%, of the 26.5 percent decline in stayer wage-change dispersion; negotiation history contributes 6.2 points. Without the U-shape moments, negotiation history instead contributes 16.7 points, or about 94%, of the 17.8 percent decline, while learning contributes only 3.0 points (95% CI: [1.3, 4.7]). Other lifecycle forces offset the decline by 2.5 and 2.0 points, respectively. Thus, the same familiar decline in wage volatility (Meghir and Pistaferri, 2004; Guvenen, 2009; Altonji, Smith Jr. and Vidangos, 2013) arises from different economic forces across the two estimates. By disciplining the persistence of innovations in match productivity and the evolution of beliefs, the U-shape moments select learning as the principal force in the preferred lifecycle profile. The reversal is concentrated among stayers: both estimates generate similar rising wage-change dispersion among job-to-job movers, consistent with job shopping as a source of large wage changes (Bagger et al., 2014); Online Appendix Figure OA.8 reports these profiles.

7.2 Learning, Misallocation, and Output

In the tradition of Jovanovic (1979), learning affects allocative efficiency by changing which matches workers retain and which outside offers they accept. We compare each estimated learning economy with its full-information benchmark to quantify these mobility and output consequences.

Full-information benchmark. We use two full-information comparisons. First, we apply the full-information decision rule to the states and offers realized in the learning economy to classify erroneous and missed moves. Second, we solve and simulate the full-information economy under matched worker-period shocks to measure the effects on employment and output. In both exercises, ability is effectively known at labor-market entry ($\sigma_{a,0} = 10^{-4}$), current match productivity is observed at the post-output decision node, and each estimate is compared with its own full-information benchmark.

Let $\tilde{y}_{it} \equiv \mu_{j(i,t)} + a_i + m_{it}$ denote realized log output. We measure aggregate level output as

$$Y \equiv \mathbb{E}[\exp(\tilde{y}_{it})\mathbf{1}\{E_{it} = 1\}].$$

We classify an accepted outside offer as an *erroneous move* when the full-information rule applied to the same state and offer would reject it. A *missed move* is an offer rejected under learning that the corresponding full-information rule would accept.

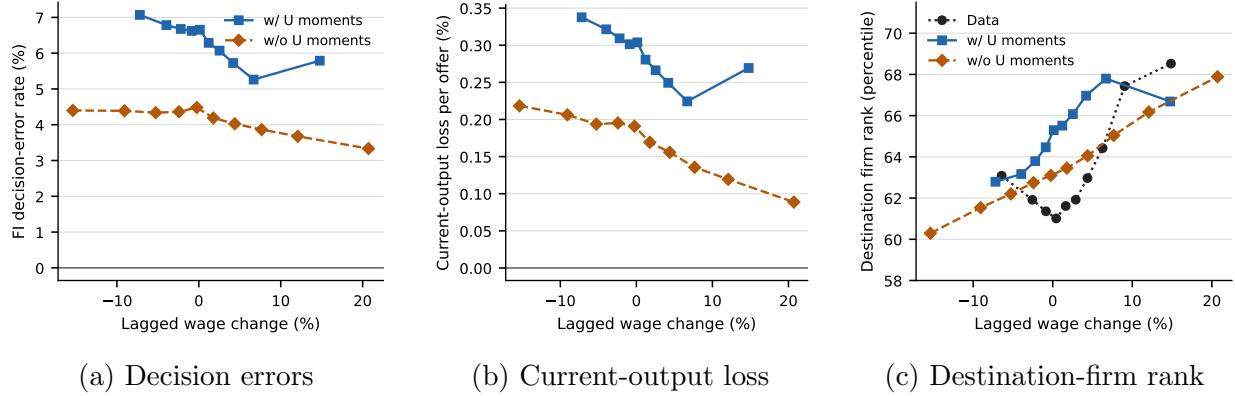
Table 6: Learning relative to full information

	With U moments	Without U moments
Erroneous moves (%)	3.32	2.21
Missed moves (%)	3.60	2.23
Employment gap (p.p.)	0.013	-0.014
Aggregate level-output gap (%)	-0.229	-0.073

Notes: Erroneous moves and missed moves are percentages of on-the-job offer decisions. An erroneous move is accepted under learning but rejected by the corresponding FI rule; a missed move is rejected under learning but accepted under FI. These policy comparisons apply the FI rule to states and offers realized in the learning economy. The employment and aggregate level-output gaps, reported as learning minus FI, compare separately solved and simulated equilibria under common worker-period shocks and the same parameter vector apart from the information structure. Online Appendix Table [OA.32](#) reports delta-method uncertainty under both estimates.

Aggregate effects. Table 6 shows that learning and full-information rules disagree for 6.92 percent of on-the-job offer decisions under the preferred estimate: 3.32 percent are erroneous moves and 3.60 percent are missed moves. The corresponding rate is 4.44 percent without the U-shape moments. The preferred learning economy produces 0.229 percent less aggregate level output than its full-information benchmark (95% CI: [0.214, 0.245]), compared with 0.073 percent (95% CI: [0.035, 0.110]) without the U-shape moments. Both losses are precisely positive, and the preferred point estimate is approximately three times as large. Net employment gaps remain small: 0.013 percentage points under the preferred estimate (95% CI: [-0.005, 0.032]) and -0.014 points without the U-shape moments (95% CI: [-0.028, -0.001]). The preferred interval includes zero, whereas the no-U interval lies narrowly below zero. Beneath these net effects, learning lowers endogenous separations under the preferred estimate, adding 0.131 percentage points to employment, but raises them without the U-shape moments, subtracting 0.115 points. Differences in entry and layoff exposure offset most of either effect. Thus, the U-shape moments alter retention and the allocation of employed workers while leaving aggregate employment nearly unchanged. Online Appendix Table [OA.31](#) reports the complete accounting. As a robustness check, the extended specification with firm-dependent signal precision and worker-firm complementarity (Section 6.3) delivers virtually the same allocative implication: aggregate output loss is 0.233 percent both at the estimated γ and κ and after setting them to zero while holding the remaining parameters fixed (Online Appendix Table [OA.30](#)). This locates the aggregate loss in learning with persistent match productivity, the channel retained in the additive baseline.

Figure 6: Information and reallocation across the wage-change distribution



Notes: The figure averages three simulations of 500,000 workers. Model observations are divided into estimate-specific deciles of lagged wage changes and plotted at each decile's mean wage change. Blue squares report the preferred estimate, orange diamonds the reestimation that omits the U-shape moments, and black circles in panel (c) the data using the empirical decile cutoffs. Panel (a) reports erroneous moves plus missed moves as a percentage of employed workers receiving an outside offer. Panel (b) reports the contemporaneous level-output loss from learning relative to the full-information decision, scaled by mean output among employed workers. Panel (c) reports the mean percentile rank of the destination firm among job-to-job movers. Model comparisons use common worker-period shocks; the data use the corrected origin-job timing convention.

Allocation across the wage-change distribution. Aggregate output losses conceal where imperfect information affects mobility. Figure 6 therefore groups on-the-job offer decisions by the worker's preceding wage change in the learning economy. It reports the total rate of full-information decision errors, their current-output cost, and the destination-firm rank of workers who move. This organization connects the allocation exercise to the preceding wage changes used to construct the U-shape moments.

The preferred estimate produces a distinctive pattern in the upper tail. Its total decision-error rate falls from 7.07 percent in the first wage-change decile to 5.26 percent in the ninth, but rises to 5.79 percent in the tenth. The associated current-output loss falls from 0.338 to 0.224 percent of mean employed output and then rises to 0.269 percent. The ninth-to-tenth-decile increases are 0.528 percentage points for decision errors (95% CI: [0.325, 0.730]) and 0.0449 percentage points of mean employed output (95% CI: [0.0294, 0.0604]). Without the U-shape moments, both objects continue to decline through the top decile. From the ninth to the tenth decile, the decision-error rate falls by 0.342 percentage points (95% CI: [-0.390, -0.294]), and current-output loss falls by 0.0306 percentage points (95% CI: [-0.0413, -0.0200]). Thus, the specification-specific intervals place both upper-tail changes above zero under the preferred estimate and below zero when the U-shape moments are omitted. Large wage gains nevertheless represent productive upgrading in both estimated economies: among

job-to-job movers following wage gains, destination-firm rank exceeds origin-firm rank by 20.93 percentile points under the preferred estimate and by 20.40 points without the U-shape moments. In the ninth and tenth deciles, destination firms lie at the 67.8th and 66.7th percentiles under the preferred estimate, at the 66.2nd and 67.9th percentiles without the U-shape moments, and at the 67.4th and 68.5th percentiles in the data. Thus, the right branch of the U-shape combines movement toward high-ranked firms with a rebound in information-induced departures from the full-information rule and output losses. The no-U estimate reproduces upgrading; the U-shape moments additionally select an upper-tail rebound in information-induced misallocation.

The mechanism applies the moving-target insight of [Kahn and Lange \(2014\)](#) to match productivity. Whereas evolving worker productivity sustains uncertainty in their analysis, persistent match productivity here slows inference about fixed worker ability and makes information consequential for mobility. The near-zero estimates of γ and κ show that this allocative channel does not rely on worker–firm complementarity or cross-firm variation in signal precision.

These economic conclusions are also robust to the relative weight on the U-shape moments. Halving their contribution to the estimation loss produces nearly the same total decline in stayer wage-change dispersion as omitting them (18.2 versus 17.8 percent), yet learning accounts for 97 rather than 17 percent of that decline. Doubling the contribution leaves the learning share at 77 percent, compared with 86 percent in the preferred specification. The corresponding aggregate output losses are 0.191 and 0.179 percent, each more than twice the 0.073 percent loss obtained when the U-shape moments are omitted. Online Appendix Table [OA.24](#) reports the full comparison.

What the U-shape adds. The two estimates generate the familiar decline in stayer wage-change volatility through different mechanisms. The structural decomposition assigns most of the decline to learning under the preferred estimate and almost all of it to accumulated negotiation capital when the U-shape moments are omitted. The U-shape moments also select persistent match risk that produces upper-tail rebounds in decision errors and output losses, changes the point-estimated endogenous-separation response from excess separation to excess retention, and raises the point-estimated aggregate output loss from imperfect information by approximately a factor of three. Net employment effects remain close to zero. The U-shape therefore links the wage-change pattern used in estimation to a distinct account of career wage dynamics and to quantitatively different implications for retention and reallocation.

8 Conclusion

This paper documents a U-shaped relationship between within-job wage changes and subsequent separations in administrative data from Italy, Germany, and Austria. In Italy, most of the curvature lies within tenure and is accounted for by job-to-job transitions. The U-shape therefore contains information beyond the familiar lifecycle declines in wage volatility and mobility.

We use this information to estimate a random-search model on Italian data. Both the preferred estimate and a reestimation that omits the U-shape moments reproduce declining within-job wage-change volatility, but through different economic mechanisms. Under the preferred estimate, learning accounts for 86% of the decline and match productivity has a half-life of 14.5 years. In the reestimation, accumulated negotiation history accounts for 94% and the half-life falls to 3.5 years. Offer acceptance and job-to-job mobility rise among workers with the largest wage gains under the preferred estimate but continue to decline in the reestimation. Relative to their respective full-information benchmarks, imperfect information lowers aggregate output by 0.229% and 0.073%.

Similar lifecycle profiles of wage-change volatility can arise from economically distinct earnings processes, with different implications for persistence, mobility, and output. Employment transitions are informative about these differences because information about worker ability and match productivity affects both wage revisions and the value of reallocation. The joint behavior of earnings and mobility therefore provides information about both the persistence of idiosyncratic earnings risk and its allocative consequences.

A Theoretical Derivations

A.1 Learning Benchmark: Wage-Change Variance

Output in the benchmark is

$$y_{i,j,t} = \mu_j + a_i + \eta_{i,j,t}, \quad \eta_{i,j,t} \sim \mathcal{N}(0, \sigma_\eta^2),$$

with prior $a_i \sim \mathcal{N}(A_0, \sigma_{a,0}^2)$. Let $\tilde{y}_t \equiv y_t - \mu_j$ and define $A_t \equiv \mathbb{E}[a_i \mid \mathcal{I}_t]$ and $\sigma_{a,t}^2 \equiv \text{Var}(a_i \mid \mathcal{I}_t)$. Bayesian updating gives

$$A_t = A_{t-1} + K_{t-1}(\tilde{y}_t - A_{t-1}), \quad K_{t-1} = \frac{\sigma_{a,t-1}^2}{\sigma_{a,t-1}^2 + \sigma_\eta^2}, \quad (\text{A.1})$$

$$\sigma_{a,t}^2 = \frac{\sigma_{a,0}^2}{1 + t(\sigma_{a,0}^2/\sigma_\eta^2)}. \quad (\text{A.2})$$

Writing $s_t \equiv \sigma_{a,t}^2 / \sigma_\eta^2$ yields $s_t = s_0 / (1 + s_0 t)$.

Under the spot-wage normalization, the incumbent wage is set before current output is observed: $w_t = \mu_j + A_{t-1}$. Hence $\Delta w_t = A_{t-1} - A_{t-2}$ for $t \geq 2$, and the period- $(t-1)$ signal generates the period- t wage revision.

Proposition A.1 (Signal-driven wage revisions). *Before conditioning on retention after the period- $(t-1)$ signal,*

$$\Delta w_t = K_{t-2}((a_i - A_{t-2}) + \eta_{t-1}),$$

and

$$V_t^\Delta \equiv \text{Var}(\Delta w_t | t) = \sigma_\eta^2 \frac{s_{t-2}^2}{1 + s_{t-2}} = \frac{\sigma_\eta^2 s_0^2}{(1 + s_0(t-2))(1 + s_0(t-1))}. \quad (\text{A.3})$$

Thus V_t^Δ decreases strictly with accumulated signals, and hence with experience, and converges to zero.

The result follows by applying equation (A.1) to the period- $(t-1)$ signal and using $\text{Var}(a_i - A_{t-2} + \eta_{t-1} | \mathcal{I}_{t-2}) = \sigma_{a,t-2}^2 + \sigma_\eta^2$. Endogenous retention can alter the variance among observed stayers; the estimated model incorporates this selection directly.

A.2 Composition-Induced U-Shape

Let $X \equiv \Delta w_{t-1}$, let $Y \equiv \mathbf{1}\{\text{separate in period } t\}$, and let G denote the worker's tenure group. Conditional on $G = g$, suppose that X is symmetric, $\text{E}[X | G = g] = 0$, $\text{Var}(X | G = g) = V_g$, and $\text{Cov}(X^2, Y | G = g) = 0$. Let $p_g \equiv \text{Pr}(Y = 1 | G = g)$ and let $\pi_g \equiv \text{Pr}(G = g)$ denote the tenure shares. The learning benchmark implies declining wage-change dispersion before selection; the proposition below requires the corresponding selected variance V_g to decline across tenure groups. In the benchmark, an outside offer is accepted when its firm-productivity draw μ' exceeds current firm productivity μ . Hence,

$$p_g = \delta + (1 - \delta)(p_g^{EU} + \lambda_1 \ell_g), \quad \ell_g \equiv \text{E}[1 - F_\mu(\mu) | G = g], \quad (\text{A.4})$$

so selection toward more productive firms also tends to lower separation across tenure groups.

Proposition A.2 (Composition-induced U-shape). *In the pooled population regression $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$,*

$$\beta_2 = \frac{\text{Cov}(X^2, Y)}{\text{Var}(X^2)} = \frac{\text{Cov}_\pi(V_g, p_g)}{\text{Var}(X^2)}.$$

If V_g and p_g are both weakly decreasing in g and nondegenerate, then $\beta_2 > 0$.

Conditional symmetry gives $\text{Cov}(X, X^2) = 0$. The law of total covariance and conditional orthogonality then imply $\text{Cov}(X^2, Y) = \text{Cov}_\pi(V_g, p_g) > 0$. For the tenure-standardized change $Z = X/\sqrt{V_G}$, the corresponding quadratic coefficient is zero because $\text{E}[Z^2 | G] = 1$. Online Appendix OA.1.3 provides the exact Frisch–Waugh–Lovell decomposition used in the empirical analysis.

A.3 Learning with Match Productivity

The estimated model adds persistent match productivity:

$$y_{i,j,t} = \mu_j + a_i + m_{i,j,t}, \quad m_{i,j,t} = \rho m_{i,j,t-1} + \eta_{i,j,t}, \quad \eta_{i,j,t} \sim \mathcal{N}(0, \sigma_\eta^2).$$

A new match begins at the known state $m_{i,j,0} = 0$. Thus $\hat{m}_{0|0} = 0$, $P_{m,0|0} = C_{0|0} = 0$, and before the first post-hire output observation $P_{m,1|0} = \sigma_\eta^2$.

Let $\theta_t = (m_t, a_i)'$, $H = (1, 1)$, $\Phi = \text{diag}(\rho, 1)$, and $Q = \text{diag}(\sigma_\eta^2, 0)$. Net output and the state transition are

$$\tilde{y}_t \equiv y_t - \mu_j = H\theta_t, \quad \theta_t = \Phi\theta_{t-1} + (\eta_t, 0)'.$$

Write the filtered mean as $\hat{\theta}_{t|t} = (\hat{m}_{t|t}, A_t)'$ and its covariance as

$$P_{t|t} = \begin{bmatrix} P_{m,t} & C_t \\ C_t & P_{a,t} \end{bmatrix}.$$

For a continuing job, prediction gives

$$\hat{\theta}_{t|t-1} = \Phi\hat{\theta}_{t-1|t-1}, \quad P_{t|t-1} = \Phi P_{t-1|t-1} \Phi' + Q. \quad (\text{A.5})$$

At a job change, the posterior mean and variance of ability carry over, while $\hat{m}$, P_m , and C reset to zero before the first match innovation.

Define the innovation and its variance by

$$\xi_t = \tilde{y}_t - H\hat{\theta}_{t|t-1}, \quad \Omega_t = H P_{t|t-1} H'.$$

Because output contains no additional measurement error, the Kalman update is

$$K_t = P_{t|t-1} H' \Omega_t^{-1}, \quad \hat{\theta}_{t|t} = \hat{\theta}_{t|t-1} + K_t \xi_t, \quad P_{t|t} = (I - K_t H) P_{t|t-1}. \quad (\text{A.6})$$

The covariance recursion is deterministic. Let $q \equiv \sqrt{P_{a,0}}$ denote posterior ability uncertainty when the current job begins and let τ denote tenure. For given (ρ, σ_η^2) , equations (A.5)–(A.6) define $P_a(q, \tau)$, $P_m(q, \tau)$, and $C(q, \tau)$. Hence (q, τ) recovers the complete covariance state. When $\rho = 0$, match productivity is i.i.d. and the ability component of K_t reduces to the scalar Bayesian gain in equation (A.1).

A.4 Recursive Formulation of the Estimated Model

Let $J = \mu_j$ denote firm productivity, $I = (A, q)$ the worker-information state, and $M = (\hat{m}, \tau)$ the match state. Let $M^0 = (0, 0)$ be the job-start match state and let $\mathcal{G}^{I,J,M}$ denote the distribution of post-signal states (I', M') generated by the filter.

Let $V(J, I, M, R)$ be the employed worker's value when the current negotiation benchmark is R , $P(J, I, M, R)$ the firm's value, and $U(I)$ the value of non-employment. Joint surplus is

$$S(J, I, M) = P(J, I, M, R) + V(J, I, M, R) - U(I),$$

which is independent of R under transferable utility. Write $S^+ = \max\{S, 0\}$ and define the worker's feasible surplus claim by

$$b(S, R) \equiv \max\{\min\{R + \alpha(S - R), S\}, 0\}.$$

The associated values are $V = U + b(S, R)$ and $P = S - b(S, R)$.

At the start of a period, the wage is set from the pre-update state. Output is then observed and beliefs are updated. An exogenous separation occurs with probability δ . Conditional on survival, an outside offer J' arrives with probability λ_1 . With $R' = S^+(J, I', M')$ and $R'' = S(J', I', M^0)$, the worker moves or renegotiates on the sets

$$\begin{aligned}\mathcal{M}_2 &= \{J' : S(J', I', M^0) > S^+(J, I', M')\}, \\ \mathcal{M}_3 &= \{J' : R < S(J', I', M^0) < S^+(J, I', M')\}.\end{aligned}$$

The value of non-employment satisfies

$$U(I) = z + \beta U(I) + \beta \lambda_0 \alpha \int S^+(J', I, M^0) dF(J'). \quad (\text{A.7})$$

Combining worker and firm continuation values yields the joint-surplus recursion

$$\begin{aligned}S(J, I, M) &= \mathbb{E}[y_{ij} \mid J, I, M] - z \\ &+ \beta \int (U(I') - U(I)) d\mathcal{G}^{I, J, M}(I', M') \\ &- \beta \lambda_0 \alpha \int S^+(J', I, M^0) dF(J') \\ &+ \beta(1 - \delta) \int \left[R' + \lambda_1 \alpha \int ((R'')^+ - R')^+ dF(J') \right] d\mathcal{G}^{I, J, M}(I', M').\end{aligned} \quad (\text{A.8})$$

For compactness, define the worker's post-signal continuation surplus as

$$\begin{aligned}\mathcal{C}(J, I', M', R) &= \lambda_1 \left[\int_{\mathcal{M}_2} (R' + \alpha(S(J', I', M^0) - R')) dF(J') \right. \\ &\quad \left. + \int_{\mathcal{M}_3} (R'' + \alpha(S(J, I', M') - R'')) dF(J') \right] \\ &+ \left(1 - \lambda_1 \int_{\mathcal{M}_2 \cup \mathcal{M}_3} dF(J') \right) b(S(J, I', M'), R).\end{aligned}$$

The wage that implements the bargaining allocation is therefore

$$\begin{aligned} w(J, I, M, R) &= b(S(J, I, M), R) + U(I) \\ &\quad - \beta \int U(I') d\mathcal{G}^{I,J,M}(I', M') \\ &\quad - \beta(1 - \delta) \int \mathcal{C}(J, I', M', R) d\mathcal{G}^{I,J,M}(I', M'). \end{aligned} \quad (\text{A.9})$$

Equations (A.6), (A.7), (A.8), and (A.9), together with the mobility sets above, characterize beliefs, allocations, and wages in the estimated economy.

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Online Appendix to:
Separations on the Job Ladder

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Contents

OA.1	Additional Details: Empirics	47
OA.1.1	Data Appendix	47
OA.1.2	Empirical Appendix	48
OA.1.3	Exact between–within decomposition	73
OA.1.4	Decomposition Results for Germany and Austria	74
OA.1.5	Alternative Career-Stage Groupings	74
OA.2	Additional Details: Estimation	77
OA.2.1	Estimator, Sample, and Reporting Units	77
OA.2.2	Estimation Sample Comparison	77
OA.2.3	Moment Definitions	78
OA.2.4	Ten-Bin Levels and Support	79
OA.2.5	Estimate without U-Shape Moments	80
OA.2.6	Fit under Reestimated Restrictions	82
OA.2.7	Empirical Moment Uncertainty	82
OA.2.8	Local Parameter Inference	83
OA.2.9	Local Moment–Parameter Sensitivity	84
OA.2.10	Curvature under Fixed-Parameter Restrictions	84
OA.2.11	Untargeted Tenure Profiles	85
OA.3	Firm-Dependent Learning and Worker–Firm Complementarity	87
OA.3.1	Extended Model	87
OA.3.2	Matched Fit Comparison	87
OA.3.3	Allocative Implications	88
OA.4	Additional Application Results	88
OA.4.1	Wage Changes of Movers and Stayers	88
OA.4.2	Employment-Flow Accounting	89
OA.4.3	Uncertainty in Application Outcomes	90

OA.1 Additional Details: Empirics

OA.1.1 Data Appendix

OA.1.1.1 Italian Veneto Worker Histories

The Veneto Worker Histories (VWH) cover all wage workers in the private sector in the provinces Treviso and Vicenza in the North-East Italian region Veneto for the period 1975–2001.²² It further contains the work history of all employer associated workers during the sampling period if those workers move out of the sampling provinces. The data set includes information on gender, age and residency of the workers. Information on the employer covers the age of the employer, the location of the seat of the employer and the sector of its economic activity. For each worker, the VWH contains information on all employment spells during the year, including total real earnings and time worked at the job as well as the qualification at the job (worker or manager, for instance) and the nature of the job (temporary or with undetermined duration) as well as the contract type. Job qualification provides the available proxy for education (Galizzi and Lang, 1998). We construct real daily earnings by dividing total real spell earnings by days worked. Earnings include overtime payments but exclude social-insurance benefits and are recorded without a reporting ceiling. Because days worked are consistently reported only from 1982 onward, our analysis begins in 1982.

OA.1.1.2 German LIAB

The linked employer-employee dataset LIAB combines administrative records from the Federal Employment Agency (BA) with plant-level data from the Establishment Panel of the Institute of Employment Research (IAB) for the years 1993–2010.²³ For a disproportionately stratified sample of firms in operation during the years 2000–2008, over-representing large establishments, the data cover the full employment history of all employer-associated workers in the years 1999–2009 and collect work histories as far back as 1993.²⁴ East German establishments enter the panel from 1996; the 1993–1995 observations therefore cover West Germany only.²⁵ There is both an upper ceiling and a lower floor on wages subject to social security contributions in Germany. For the upper ceiling two potentially binding thresholds (one each for East and West Germany) exist, depending on the social security organization that the worker belongs to. The data set includes information on employees subject to social security contributions and

²²For descriptions of the data refer to Tattara and Valentini (2010), Leonardi and Pica (2013), Ibsen, Trevisan and Westergaard-Nielsen (2008), Grinza (2021), Bartolucci, Devicienti and Monzón (2018), Serafinelli (2019).

²³Further details on the data set can be found in Klosterhuber, Heining and Seth (2013) and Fischer et al. (2014) for a detailed description as well as Card, Heining and Kline (2013), Hirsch and Zwick (2015), Addison et al. (2008), John T. Addison (2010), Addison et al. (2015), Gürtzgen (2007).

²⁴Sampling probabilities range from 0.11% for establishments with 1–4 employees to 91.3% for establishments with 5,000 or more employees (Fischer et al., 2014).

²⁵Full coworker coverage is available only for establishments with a valid IAB survey interview in the relevant year (wave codes B/C in Fischer et al. 2014); establishments re-entering the panel after a period of exclusion receive a new identifier. Workers at non-panel firms appear in the LIAB via their employment histories but without complete coworker rosters. As a result, a full set of coworkers are only available at LIAB-sampled firms during the period 1999–2009.

excludes civil servants, family workers and students in higher education. The main variables from the data set used in this study include information on the start and end of employment spells, the type of work (temporary, full-time), the age of the worker, the wage, the sector of economic activity of the employer as well as the occupation at the three digit level. We also exploit experience and tenure variables provided as part of the data that accurately measure each workers employment histories. The LIAB is furthermore unique among the three data sets in that it offers information on education. The wage information available in the LIAB is daily gross wages up to the earnings ceiling for social security contributions. Firm-level variables in the linked Establishment History Panel (BHP) are measured as of June 30 each year and thus represent point-in-time snapshots rather than annual flow measures. We deflate wages using the CPI deflator with base year 2010.

OA.1.1.3 Austrian AMDB

The Austrian AMDB dataset is co-constructed by the federal ministry of economics and labor (BMWA) and the labor service institution AMS based on social security records.²⁶ It contains the universe of employment spells and social security benefits for Austrian workers during the time period 2000-2016 and covers start and end date, total earnings and days worked at each job for each month of the year. Moreover, the AMDB includes information on the economic sector at the 4-digit level and the work place location as well as age. The recorded income is subject to a single, nationally uniform reporting limit. We adjust observed nominal wages using the CPI deflator with base year 2010.

OA.1.2 Empirical Appendix

The following section reports further results relating to the empirical section of the paper. Section [OA.1.2.1](#) reports summary statistics. Section [OA.1.2.2](#) reports further results on the Logit specification and linear probability model specification. Section [OA.1.2.3](#) reports robustness and sensitivity checks.

OA.1.2.1 Summary Statistics

²⁶For a description of the data set, cf. [here](#). For additional information, consider [Zweimüller et al. \(2009\)](#) and [Borovičková \(2022\)](#); [Borovičková and Shimer \(2025\)](#) using a similar data set.

Online Appendix Table OA.1: Construction of Harmonized Worker-Year Panel and Attrition

	Italy		Germany		Austria	
	Dropped	Surviving	Dropped	Surviving	Dropped	Surviving
Keep i if entry age < 30	50.07	49.93	64.49	35.51	65.70	34.30
Keep i if $19 \leq \text{age} \leq 63^*$, men	49.88	25.02	40.47	21.14	49.28	17.40
Drop i if $\exp(W_t) \leq 5$ (any t) [*]	0.12	24.99	0.76	20.98	0.95	17.23
Drop i if $ \Delta W_t \geq 1$ (any t) [*]	3.76	24.06	5.84	19.75	9.86	15.53
Drop i if public sector (any t) [*]	5.30	22.78	4.08	18.95	20.05	12.42
Drop i if $\text{jobs}_t \geq 10$ (any t) [*]	0.00	22.78	0.79	18.80	0.01	12.42
Drop i if sex missing (any t) [*]	0.00	22.78	0.00	18.80	0.00	12.42

Notes: For each country, the table reports two columns. The first reports the share of the remaining sample removed by each restriction; the second reports the cumulative share of the initial sample that survives through that restriction. The sample (already at the worker-year main spell level) is sequentially restricted as listed, implementing the selection procedure of [Kline, Saggio and Sølvesten \(2020\)](#) (selection steps marked by ^{*}), plus additional restrictions on age at entry and gender. The first restriction retains individuals who enter the labor market before age 30. The second further restricts the sample to individuals aged 19–63 and to males. The third drops individuals who ever receive very low wages (daily wages $\leq 5\text{€}$). The fourth removes individuals who ever experience absolute log wage changes greater than one between consecutive years. The fifth excludes individuals who ever work in the public sector. The sixth drops individuals with ten or more jobs within a year. The seventh excludes individuals with missing gender information. Following [Kline, Saggio and Sølvesten \(2020\)](#), all restrictions qualified by (any t) are applied at the individual level: if a condition is met in any year, the individual is entirely removed from the sample. Cumulative survival shares are the running product of the restriction-specific retention rates and are therefore computed on the same basis as the reported exclusion rates. The restrictions shown here construct the harmonized worker-year panel of Section 2.2; the further restrictions defining the empirical analysis sample (Section 2.3), beginning with the removal of apprentices, are applied subsequently and are not reflected in the shares reported here.

Online Appendix Table OA.2: Empirical Analysis Sample and Attrition

	Italy		Germany		Austria	
	Dropped	Surviving	Dropped	Surviving	Dropped	Surviving
Drop <i>it</i> if $\text{apprentice}_t = 1$	5.58	94.42	0.17	99.83	N/A	100.00
Keep <i>it</i> if full-year work in $t - 1$ and $t - 2$	56.49	41.08	56.61	43.32	52.88	47.12
Drop <i>it</i> if ΔW_{t-1} undefined (< 2 yrs at firm)	1.39	40.51	3.03	42.01	1.49	46.42
Drop <i>it</i> if t is last year i is observed	11.59	35.82	9.88	37.86	10.37	41.61
Drop <i>it</i> if t is last year firm is observed	4.34	34.26	6.29	35.47	1.69	40.90
Drop <i>it</i> if consecutive job change (keep likelier t)	2.39	33.44	2.86	34.46	2.93	39.70
Drop <i>it</i> if job-change indicator missing	0.22	33.37	1.68	33.88	0.35	39.56
Drop <i>it</i> if ΔW_t outside $[p_2, p_{98}]$	4.00	32.04	4.00	32.52	4.00	37.98

Notes: For each country, the table reports two columns. The first reports the share of the remaining sample removed by each restriction; the second reports the cumulative share of the base sample that survives through that restriction. The base sample (100%) is the harmonized worker-year main-spell panel of Section 2.2, which implements the selection procedure of Kline, Saggio and Sølvesten (2020) together with additional restrictions on age at entry and gender. The restrictions reported below are applied sequentially in the order listed and define the empirical analysis sample of Section 2.3. With the exception of the first, they operate at the worker-year level: only the observations satisfying the condition are removed, while the individual continues to contribute her remaining years. The first restriction removes apprentices; it is not applied to Austria, where apprentices are already excluded from the raw data, so the Austrian column enters the subsequent restrictions unchanged. The second retains only observations preceded by two consecutive years of full-year employment, and the third requires a lagged wage change to be defined, which implies at least two years with the same employer. The fourth and fifth remove observations in the last year in which the worker, respectively the firm, is observed, since a job change cannot be measured thereafter. The sixth resolves consecutive job changes by retaining the year in which the separation is more likely to have occurred, based on days worked. The seventh drops observations for which the job-change indicator cannot be constructed, and the eighth trims the top and bottom two percent of the wage-change distribution. Cumulative survival shares are the running product of the restriction-specific retention rates and are therefore computed on the same basis as the reported exclusion rates.

Online Appendix Table OA.3: Summary Statistics by Country: Empirical Analysis Samples

	Italy			Germany			Austria		
	Mean	SD	Median	Mean	SD	Median	Mean	SD	Median
Age	31.47	5.37	31.00	33.17	5.16	33.00	32.23	5.14	32.00
Tenure at the employer	7.18	3.88	6.00	6.98	3.76	6.00	7.20	3.72	6.00
Tenure at the employer (IAB)				8.57	5.13	7.40			
Labor market experience	9.66	4.31	9.00	9.38	4.04	9.00	9.59	3.86	9.00
Labor market experience (IAB)				11.41	5.16	10.80			
Employer size	9.46	37.79	2.85				14.90	82.77	2.67
Employer size (TV & VI)	10.24	38.83	3.07						
Employer size (IAB)				207.60	860.52	41.97			
% change in log real wage	2.94	5.89	2.22	1.95	5.22	1.34	2.75	6.00	1.83
% separations	7.43	26.23	0.00	8.05	27.21	0.00	8.10	27.29	0.00
% separations into unemployment (E2U)	0.96	9.76	0.00	1.03	10.10	0.00	1.94	13.80	0.00
% separations into employment (E2E)	6.47	24.60	0.00	6.82	25.21	0.00	6.16	24.04	0.00
Obs.	1,645,579			1,229,924			2,758,316		

Notes: The table reports worker-year summary statistics for the three country samples. Entries labeled (IAB) refer to administratively provided German measures. Employer size is measured at the level of the available employer identifier. Employer size (TV & VI) reports employer size only for employers in Treviso and Vicenza, where the universe of employment relationships are observed. EU transitions encompass all separations that are followed by an unemployment spell of more than one month, EE transitions are the complement.

Online Appendix Table OA.4: Sample Composition by Country: Empirical Analysis Samples

	Italy		Germany		Austria	
	Count	Mean	Count	Mean	Count	Mean
Workers	306,057		216,633		452,273	
Employers	68,129		82,231		93,088	
Manufacturing (workers)		56.30		54.29		38.82
Manufacturing (employers)		44.35		36.83		19.51
% Blue collar		68.78				
% White collar		30.31				
% Secondary-intermediary degree				87.88		
% University degree				9.83		
Obs.	1,645,579		1,229,924		2,758,316	

Notes: The table reports sample-composition statistics for the three country samples. Share variables are reported in percent.

Online Appendix Table OA.5: Separation Rates by Age

	Italy		Germany		Austria	
	Mean	SD	Mean	SD	Mean	SD
Age 25						
% separations	10.64	30.84	9.99	29.99	13.45	34.12
Obs.	88,778		34,887		130,536	
Age 35						
% separations	5.64	23.06	7.91	26.99	7.53	26.39
Obs.	74,560		71,541		154,562	
Age 45						
% separations	4.13	19.90	4.88	21.54	3.79	19.09
Obs.	9,899		8,246		12,387	

Notes: The table reports mean separation rates and their standard deviations by age group (25, 35, and 45) and country. Reported values are expressed in percentage points.

OA.1.2.2 Further Results and Figures

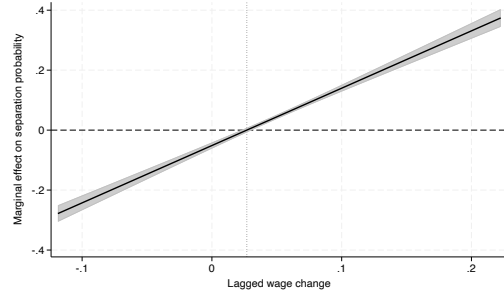
Online Appendix Table OA.6: Logit Estimates: Timing and Measurement of Wage Changes

	Italy			Germany			Austria		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1st decile	1.20*** (0.016)	1.40*** (0.020)	1.34*** (0.017)	1.24*** (0.018)	1.84*** (0.027)	1.48*** (0.022)	1.40*** (0.014)	1.88*** (0.019)	1.64*** (0.016)
2nd decile	1.07*** (0.014)	1.11*** (0.017)	1.18*** (0.016)	1.14*** (0.017)	1.51*** (0.023)	1.37*** (0.021)	1.18*** (0.012)	1.16*** (0.013)	1.44*** (0.014)
3rd decile	1.04** (0.014)	1.01 (0.016)	1.12*** (0.015)	1.12*** (0.017)	1.22*** (0.019)	1.35*** (0.020)	1.09*** (0.011)	1.14*** (0.013)	1.35*** (0.014)
4th decile	1.00 (.)	1.00 (.)	1.07*** (0.015)	1.17*** (0.018)	1.26*** (0.020)	1.29*** (0.020)	1.05*** (0.011)	1.10*** (0.012)	1.18*** (0.012)
5th decile	1.01 (0.014)	1.00 (0.016)	1.03* (0.014)	1.05*** (0.016)	1.07*** (0.017)	1.14*** (0.018)	1.00 (.)	1.00 (.)	1.07*** (0.011)
6th decile	1.04* (0.014)	1.10*** (0.017)	1.00 (.)	1.03* (0.016)	1.00 (.)	1.05** (0.017)	1.00 (0.010)	1.15*** (0.013)	1.00 (.)
7th decile	1.03* (0.014)	1.24*** (0.019)	1.00 (0.014)	1.01 (0.016)	1.01 (0.016)	1.00 (.)	1.02 (0.010)	1.19*** (0.013)	1.02 (0.011)
8th decile	1.05*** (0.014)	1.47*** (0.021)	1.01 (0.014)	1.00 (.)	1.04* (0.017)	1.00 (0.016)	1.04*** (0.011)	1.35*** (0.015)	1.06*** (0.011)
9th decile	1.06*** (0.014)	1.87*** (0.026)	1.02 (0.014)	1.03 (0.016)	1.16*** (0.018)	1.06*** (0.017)	1.09*** (0.011)	1.62*** (0.017)	1.11*** (0.012)
10th decile	1.10*** (0.015)	3.45*** (0.044)	1.10*** (0.015)	1.21*** (0.018)	1.58*** (0.024)	1.23*** (0.019)	1.13*** (0.011)	2.61*** (0.025)	1.18*** (0.012)
Obs.	1,645,579	1,645,579	1,645,579	1,229,924	1,229,924	1,229,924	2,758,316	2,758,316	2,758,316
Wage-change measure	Observed	Observed	Residual	Observed	Observed	Residual	Observed	Observed	Residual
Wage-change timing	$t-1$	t	$t-1$	$t-1$	t	$t-1$	$t-1$	t	$t-1$

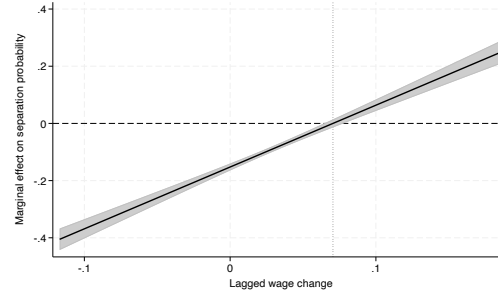
Notes: The table reports odds ratios from logit regressions of separation in year t on deciles of wage changes. Columns (1), (2), (4), (5), (7), and (8) use observed wage changes, while columns (3), (6), and (9) use residualized wage changes. Columns (1), (3), (4), (6), (7), and (9) use lagged wage changes, whereas columns (2), (5), and (8) use contemporaneous wage changes. The omitted category is the lowest separation hazard wage-change bin. Residualized wage changes are obtained after controlling for year fixed effects, lagged tenure, lagged experience, lagged age, industry dummies and where available occupation dummies. Standard errors are reported in parentheses.* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Online Appendix Figure OA.1: Marginal Effect of Wage Changes on Subsequent Separations

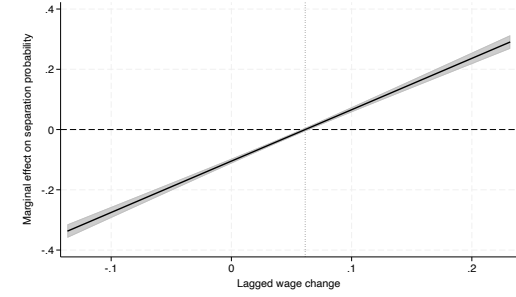
Baseline



(a) Italy

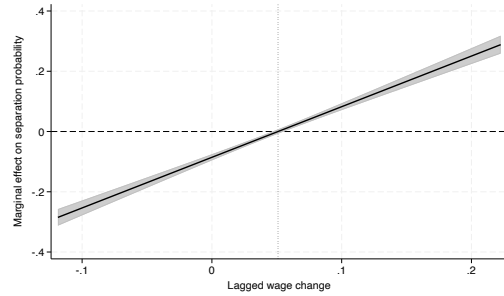


(b) Germany

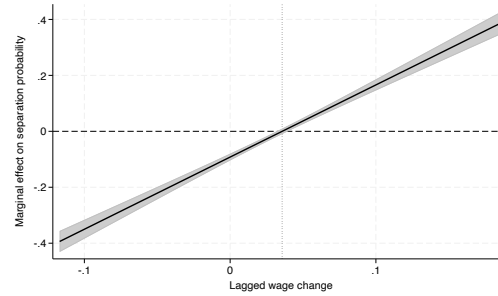


(c) Austria

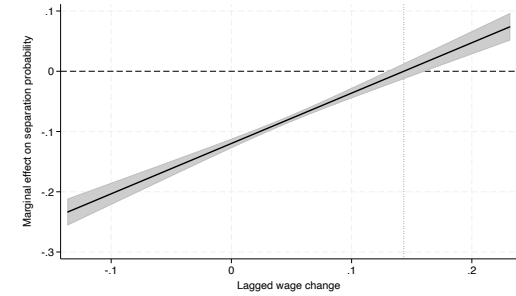
Augmented



(d) Italy



(e) Germany



(f) Austria

Notes: The figure plots the marginal effect of the lagged wage change on the probability of separation. For the quadratic specification, the marginal effect at wage change Δw is $\hat{\beta}_1 + 2\hat{\beta}_2\Delta w$. The solid line reports the estimated marginal effect and the shaded region shows the 95% pointwise confidence interval constructed from the joint variance-covariance matrix of the linear and quadratic coefficients. The horizontal dashed line marks a marginal effect of zero. The vertical dotted line marks the implied trough, $-\hat{\beta}_1/(2\hat{\beta}_2)$. The top row includes lagged wages only. The bottom row additionally controls for lagged experience, tenure, employer size, and employer-size growth. Each panel is evaluated over the observed support of lagged wage changes in the corresponding regression sample.

Online Appendix Table OA.7: Implied Trough of the Wage-Change–Separation Relationship

	Italy		Germany		Austria	
	(1)	(2)	(3)	(4)	(5)	(6)
Implied trough (log points)	2.69	5.10	3.57	7.05	6.14	14.34
95% confidence interval	[2.29, 3.09]	[4.67, 5.52]	[3.19, 3.95]	[6.40, 7.71]	[5.80, 6.48]	[12.80, 15.88]
Percentile of ΔW_{t-1} distribution	53.7	69.5	67.5	84.5	76.7	95.1
Share to the right of trough (%)	46.3	30.5	32.5	15.5	23.3	4.9
Obs.	1,645,579	1,645,579	1,229,924	1,229,924	2,758,316	2,758,316

Notes: The table reports the trough implied by quadratic linear probability models of separation in year t on the lagged wage change, ΔW_{t-1} , its square, and lagged log wages. Columns (1), (3), and (5) report the specification with lagged log wages. Columns (2), (4), and (6) additionally control for lagged experience, tenure, employer size, and employer-size growth. The implied trough is calculated as $-\hat{\beta}_1/(2\hat{\beta}_2)$ and is expressed in log points. The reported 95% confidence intervals are calculated using the delta method. The percentile is the empirical percentile of the estimated trough in the corresponding regression sample. The share to the right is the percentage of observations for which ΔW_{t-1} exceeds the estimated trough.

Online Appendix Table OA.8: Marginal Effect of Wage Changes at Selected Points

	Italy		Germany		Austria	
	(1)	(2)	(3)	(4)	(5)	(6)
-10 log points	-0.243*** [-0.267, -0.218]	-0.254*** [-0.278, -0.229]	-0.350*** [-0.384, -0.316]	-0.369*** [-0.402, -0.335]	-0.275*** [-0.293, -0.257]	-0.203*** [-0.222, -0.185]
+10 log points	0.140*** [0.128, 0.151]	0.082*** [0.071, 0.094]	0.166*** [0.146, 0.186]	0.064*** [0.044, 0.084]	0.066*** [0.057, 0.074]	-0.036*** [-0.045, -0.028]
+15 log points	0.235*** [0.217, 0.254]	0.166*** [0.148, 0.185]	0.295*** [0.264, 0.326]	0.172*** [0.140, 0.203]	0.151*** [0.138, 0.164]	0.006 [-0.008, 0.019]
+20 log points	0.331*** [0.305, 0.357]	0.250*** [0.224, 0.277]	0.424*** [0.381, 0.467]	0.280*** [0.236, 0.323]	0.236*** [0.217, 0.255]	0.047*** [0.028, 0.066]
Obs.	1,645,579	1,645,579	1,229,924	1,229,924	2,758,316	2,758,316

Notes: The table reports the local slope of the fitted quadratic relationship between lagged wage changes and subsequent separation. At a wage change Δw_0 , the reported slope is $\hat{\beta}_1 + 2\hat{\beta}_2\Delta w_0$. Columns (1), (3), and (5) report the specification with lagged log wages. Columns (2), (4), and (6) additionally control for lagged experience, tenure, employer size, and employer-size growth. Two-sided 95% confidence intervals, calculated from the joint robust variance-covariance matrix of the linear and quadratic coefficients, are reported in brackets. A confidence interval entirely below zero at -10 log points supports a declining left arm. A confidence interval entirely above zero at a positive wage change supports an increasing right arm. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Online Appendix Table OA.9: Wage Changes and Subsequent Separations: Linear Probability Model with Linear Term Only

	Italy		Germany		Austria	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔW_{t-1}	-0.015*** (0.0036)	-0.012** (0.0036)	-0.028*** (0.0049)	-0.086*** (0.0050)	-0.058*** (0.0029)	-0.084*** (0.0029)
W_{t-1}		-0.028*** (0.00077)		0.0029** (0.00096)		-0.041*** (0.00058)
Exp_{t-1}		0.00033*** (0.000067)		0.00019* (0.000084)		0.0018*** (0.000062)
Ten_{t-1}		-0.0056*** (0.000070)		-0.0074*** (0.000088)		-0.0091*** (0.000062)
Obs.	1,645,579	1,645,579	1,229,924	1,229,924	2,758,316	2,758,316

Notes: The table reports coefficients from linear probability models of separation in year t on lagged wage changes. Columns (1), (3), and (5) have no controls. Columns (2), (4), and (6) control for lagged log wage, lagged experience, lagged tenure, lagged employer size, and lagged employer-size growth. Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Online Appendix Table OA.10: Wage Changes and Subsequent Separations: Flexible Career-Stage Controls

	(a)	(b)	(c)	(d)	(e)	(f)
Panel A: Italy						
ΔW_{t-1}	-0.051*** (0.0052)	-0.086*** (0.0052)	-0.088*** (0.0052)	-0.090*** (0.0053)	-0.093*** (0.0053)	-0.094*** (0.0053)
ΔW_{t-1}^2	0.96*** (0.042)	0.84*** (0.041)	0.83*** (0.042)	0.84*** (0.042)	0.82*** (0.042)	0.81*** (0.042)
Minimum (log points)	2.69 (0.20)	5.10 (0.22)	5.31 (0.22)	5.32 (0.22)	5.65 (0.23)	5.86 (0.24)
Slope at +10 log points	14.0 (0.58)	8.24 (0.58)	7.80 (0.58)	7.89 (0.58)	7.14 (0.58)	6.67 (0.58)
Share right of minimum (%)	46.3	30.5	29.3	29.3	27.6	26.6
Obs.	1,645,579	1,645,579	1,645,579	1,645,579	1,645,579	1,645,579
Panel B: Germany						
ΔW_{t-1}	-0.092*** (0.0062)	-0.15*** (0.0062)	-0.15*** (0.0062)	-0.15*** (0.0063)	-0.16*** (0.0062)	-0.16*** (0.0062)
ΔW_{t-1}^2	1.29*** (0.063)	1.08*** (0.063)	1.03*** (0.063)	1.13*** (0.063)	1.07*** (0.063)	1.06*** (0.063)
Minimum (log points)	3.57 (0.19)	7.05 (0.33)	7.49 (0.37)	6.84 (0.31)	7.49 (0.36)	7.69 (0.37)
Slope at +10 log points	16.6 (1.02)	6.37 (1.02)	5.15 (1.03)	7.17 (1.02)	5.37 (1.02)	4.88 (1.02)
Share right of minimum (%)	32.5	15.5	14.1	16.3	14.1	13.5
Obs.	1,229,924	1,229,924	1,229,924	1,229,924	1,229,924	1,229,924
Panel C: Austria						
ΔW_{t-1}	-0.10*** (0.0040)	-0.12*** (0.0040)	-0.12*** (0.0041)	-0.15*** (0.0042)	-0.16*** (0.0042)	-0.16*** (0.0042)
ΔW_{t-1}^2	0.85*** (0.030)	0.42*** (0.030)	0.29*** (0.030)	0.54*** (0.031)	0.50*** (0.031)	0.48*** (0.031)
Minimum (log points)	6.14 (0.17)	14.3 (0.79)	20.5 (1.72)	14.2 (0.61)	15.7 (0.74)	16.7 (0.84)
Slope at +10 log points	6.58 (0.43)	-3.63 (0.44)	-6.14 (0.44)	-4.53 (0.45)	-5.73 (0.45)	-6.38 (0.45)
Share right of minimum (%)	23.3	4.9	0.9	5.1	3.7	2.9
Obs.	2,758,316	2,758,316	2,758,316	2,758,316	2,758,316	2,758,316
Exp _{t-1} (linear)	No	Yes	Yes	No	No	No
Ten _{t-1} (linear)	No	Yes	No	Yes	No	No
Tenure dummies	No	No	Yes	No	Yes	Yes
Experience dummies	No	No	No	Yes	Yes	Yes
Tenure × experience cells	No	No	No	No	No	Yes

Notes: Linear probability models of separation in year t , controlling for W_{t-1} . Columns (b)–(f) additionally control for lagged employer size and lagged employer-size growth. In (f), tenure-year and experience-year dummies are nested in the tenure × experience cells. The minimum is expressed in log points, and the slope is multiplied by 100. Parentheses report heteroskedasticity-robust standard errors for the regression coefficients and delta-method standard errors for the minimum and slope. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Online Appendix Table OA.11: Wage Changes and Subsequent Separations by Tenure Bin

	Italy	Germany	Austria
	(1)	(2)	(3)
$\Delta W_{t-1} \times \text{Tenure 2-5}$	-0.12*** (0.0082)	-0.17*** (0.0093)	-0.20*** (0.0062)
$\Delta W_{t-1} \times \text{Tenure 6-9}$	-0.11*** (0.0092)	-0.19*** (0.011)	-0.12*** (0.0070)
$\Delta W_{t-1} \times \text{Tenure 10-13}$	-0.020 (0.010)	-0.13*** (0.013)	-0.076*** (0.0083)
$\Delta W_{t-1} \times \text{Tenure 14+}$	-0.011 (0.015)	-0.040* (0.016)	-0.025 (0.014)
$\Delta W_{t-1}^2 \times \text{Tenure 2-5}$	0.86*** (0.063)	0.94*** (0.089)	0.51*** (0.043)
$\Delta W_{t-1}^2 \times \text{Tenure 6-9}$	0.88*** (0.072)	1.28*** (0.12)	0.67*** (0.054)
$\Delta W_{t-1}^2 \times \text{Tenure 10-13}$	0.54*** (0.087)	1.69*** (0.16)	0.43*** (0.069)
$\Delta W_{t-1}^2 \times \text{Tenure 14+}$	1.04*** (0.13)	0.57** (0.18)	0.26* (0.12)
W_{t-1}	-0.028*** (0.00077)	0.0048*** (0.00096)	-0.037*** (0.00058)
Slope at +10, Tenure 2-5	5.46 (0.84)	2.33 (1.37)	-10.3 (0.60)
Slope at +10, Tenure 6-9	6.74 (0.98)	6.38 (1.96)	0.93 (0.78)
Slope at +10, Tenure 10-13	8.74 (1.32)	20.3 (2.73)	1.04 (1.07)
Slope at +10, Tenure 14+	19.7 (2.01)	7.42 (3.21)	2.61 (1.93)
p -value: equal ΔW_{t-1}^2 coefficients	0.002	<0.001	0.003
Obs.	1,645,579	1,229,924	2,758,316

Notes: Linear probability models of separation in year t with ΔW_{t-1} and ΔW_{t-1}^2 interacted with four disjoint tenure bins. All columns additionally control for lagged employer size and lagged employer-size growth, and tenure $\times$ experience cell dummies. Slopes are multiplied by 100. Parentheses report heteroskedasticity-robust standard errors for the regression coefficients and delta-method standard errors for the slopes. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

OA.1.2.3 Robustness and Sensitivity

We assess the robustness of the U-shaped relationship in six ways. First, we verify in the German data that part-time work is not driving the result. Second, we restrict the VWH to fully covered provinces only. Third, we residualize wage growth with respect to observable characteristics. Fourth, we distinguish different types of separations. Fifth, we control for workers' relative wage position within occupation. Sixth, we examine whether the pattern is consistent with a simple search-intensity interpretation. We then report two additional sensitivity checks concerning the timing convention and the East–West split in Germany.

Part-time work. Excluding part-time spells in the German data leaves the results unchanged and makes the U-shape slightly more pronounced. Online Appendix Table OA.12 (columns (1) and (2)) reports the reestimated linear probability model, in which the coefficient on squared wage growth rises from 1.29 in the baseline to 1.35. These results indicate that part-time is not an important driver of the German estimates.

Restricted VWH. Our baseline analysis uses the full Italian VWH, in line with the related literature (Kline, Saggio and Sølvesten, 2020). This is the natural benchmark because it preserves comparability with existing work and maximizes sample size. At the same time, the VWH represents the universe of employment relationships only for employers located in Treviso and Vicenza, which is relevant for our construction of employer size. We therefore reestimate the specification on the subset of employers in the fully covered provinces. As shown in Online Appendix Table OA.12 (columns (7)–(10)), the U-shaped relationship remains clearly visible and is, if anything, somewhat stronger in the restricted sample, despite a reduction of the Italian sample to about 40% of its baseline size.

Online Appendix Table OA.12: Wage Changes and Subsequent Separations: Linear Probability Model, Sample Robustness

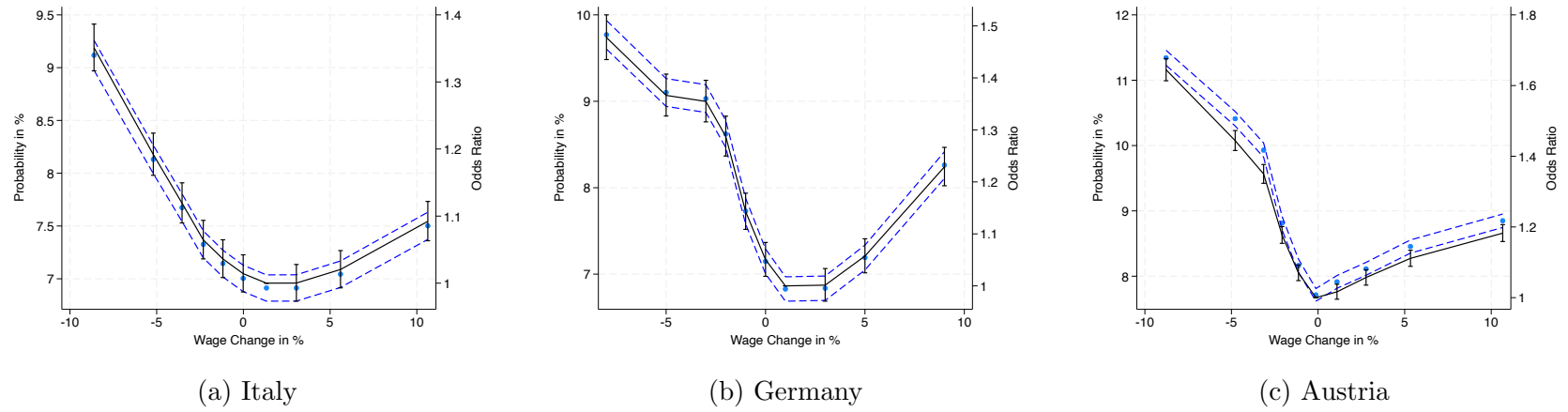
	Germany						Italy			
	Excl. part-time		West		East		Full VWH		Treviso & Vicenza	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ΔW_{t-1}	-0.100*** (0.0063)	-0.160*** (0.0063)	-0.066*** (0.0067)	-0.130*** (0.0068)	-0.200*** (0.015)	-0.290*** (0.016)	-0.051*** (0.0052)	-0.086*** (0.0052)	-0.081*** (0.0085)	-0.120*** (0.0085)
ΔW_{t-1}^2	1.350*** (0.066)	1.120*** (0.066)	1.080*** (0.069)	0.920*** (0.069)	2.220*** (0.16)	1.840*** (0.16)	0.960*** (0.042)	0.840*** (0.041)	1.080*** (0.067)	0.940*** (0.067)
W_{t-1}	-0.019*** (0.00088)	0.012*** (0.00098)	-0.022*** (0.0011)	0.013*** (0.0012)	-0.028*** (0.0020)	-0.0081*** (0.0021)	-0.061*** (0.0007)	-0.029*** (0.0008)	-0.056*** (0.0012)	-0.028*** (0.0013)
Exp_{t-1}		0.00018* (0.000085)		0.00055*** (0.000097)		-0.0036*** (0.00020)		0.00038*** (0.00007)		0.00046*** (0.00010)
Ten_{t-1}		-0.0073*** (0.000089)		-0.0072*** (0.00010)		-0.0080*** (0.00018)		-0.0056*** (0.00007)		-0.0062*** (0.00011)
Obs.	1,208,529	1,208,529	989,989	989,989	239,935	239,935	1,645,579	1,645,579	698,605	698,605

Notes: The table reports coefficients from linear probability models of separation in year t on lagged wage changes. The estimating equation is Equation (3). Columns (1)–(2) use the German sample excluding part-time workers based on LIAB information. Columns (3)–(6) report results separately for West and East Germany. Columns (7)–(10) use Italian data, where columns (9)–(10) restrict attention to the provinces of Treviso and Vicenza, for which the universe of employment relationships is observed. Odd-numbered columns include lagged log wages as controls. Even-numbered columns additionally control for lagged experience, lagged tenure, lagged employer size, and lagged employer-size growth. Robust standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Residualized wage changes. Residualizing wage growth with respect to observable worker and job characteristics leaves the U-shape intact.²⁷ Specifically, we regress wage growth on year fixed effects, lagged tenure, lagged experience, lagged age, and industry dummies. For Germany, we additionally control for three-digit occupation. Whereas including controls in the LPM removes additive correlation between observables and the separation probability, residualizing wage growth subtracts its linear projection on observables before re-estimating the wage-growth–separation relationship. Online Appendix Figure [OA.2](#) and Online Appendix Table [OA.6](#) (columns (3), (6) and (9)) show that the U-shape remains and becomes slightly more pronounced after residualization.

²⁷The robustness question is whether observed career and job characteristics account for the component of wage growth associated with subsequent separation. We therefore project wage growth directly on those characteristics. Residualizing wage levels first and then differencing would answer a different question by making the residual change depend on the wage-level specification in each year.

Online Appendix Figure OA.2: Residual Wage Changes and Subsequent Separations



Notes: The figure plots the separation rate in year t against residualized wage changes in year $t - 1$ by country. The blue dots show non-parametric within-bin separation rates (left axis, Equation (1)). The dashed lines connect the associated 95% confidence intervals, based on the standard error of the mean separation rate within each bin. The solid line and 95% confidence markers show the odds ratio of separation relative to the lowest-hazard residual wage-change bin from the logit specification (right axis, Equation (2)). Wage changes are measured in log points. Each marker corresponds to one wage-change bin in the country-specific sample and represents about 123K observations for Germany, 165K for Italy and 276K for Austria.

Types of transitions. The U-shape appears separately for employment-to-employment and employment-to-unemployment transitions (see Online Appendix Figure OA.4 for non-parametric and Online Appendix Table OA.13 for LPM results), as well as for an approximation of layoffs and quits in the German data (Online Appendix Table OA.14). One concern is that the aggregate U-shape reflects the overlay of two distinct monotone relationships: falling match productivity could simultaneously generate wage cuts and layoffs, while rising productivity could lead to voluntary quits.

To examine this possibility, we first decompose separations into employment-to-employment (E2E) transitions, defined as job changes followed by a non-employment spell of less than 32 days, and employment-to-unemployment (E2U) transitions, which involve interruptions of at least 32 days. Although this threshold does not perfectly separate quits from layoffs, it provides a consistent definition across all three datasets and distinguishes transitions with different implications for job-ladder position. Online Appendix Table OA.13 estimates the LPM separately for each transition type and shows that the U-shape remains statistically significant for both E2E and E2U separations. Online Appendix Figure OA.3 shows that the quadratic effect is sufficiently large to overcome the linear coefficient in the tails of the wage-change distribution for both types of transitions. The presence of the U-shape on both margins makes the mechanical-overlay interpretation less compelling.

We next ask whether the pattern remains when separations are restricted to likely layoffs. In Germany, unemployment benefits are suspended for the first 12 weeks following a voluntary quit but are paid immediately after an involuntary separation.²⁸ Online Appendix Table OA.14 therefore decomposes separations into those followed by benefit receipt within one month and within three months of the separation date. The U-shaped pattern remains under both layoff definitions, which suggests that it is not driven solely by voluntary quits.

²⁸Eligibility requires contributing to unemployment insurance for at least 12 of the 30 months preceding the claim. By construction, all workers in our sample satisfy this condition, so benefit receipt within the first 12 weeks is informative about involuntary separations.

Online Appendix Table OA.13: Wage Changes and Subsequent Separations by Transition Type

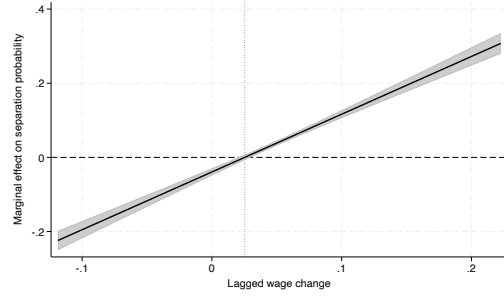
	Italy			Germany			Austria		
	Baseline (1)	E2E (2)	E2U (3)	Baseline (4)	E2E (5)	E2U (6)	Baseline (7)	E2E (8)	E2U (9)
ΔW_{t-1}	-0.051*** (0.0052)	-0.040*** (0.0049)	-0.012*** (0.0020)	-0.092*** (0.0062)	-0.058*** (0.0057)	-0.027*** (0.0026)	-0.100*** (0.0040)	-0.058*** (0.0035)	-0.047*** (0.0022)
ΔW_{t-1}^2	0.960*** (0.042)	0.780*** (0.039)	0.170*** (0.016)	1.290*** (0.063)	1.000*** (0.058)	0.240*** (0.025)	0.850*** (0.030)	0.540*** (0.026)	0.310*** (0.016)
W_{t-1}	-0.061*** (0.00070)	-0.046*** (0.00065)	-0.015*** (0.00027)	-0.027*** (0.00087)	0.001 (0.00079)	-0.025*** (0.00036)	-0.066*** (0.00055)	-0.039*** (0.00048)	-0.027*** (0.00029)
Obs.	1,645,579	1,645,579	1,645,579	1,229,924	1,229,924	1,229,924	2,758,316	2,758,316	2,758,316

Notes: The table reports coefficients from linear probability models of separation in year t on lagged wage changes and lagged log wages. Columns (1), (4), and (7) use the baseline separation indicator. Columns (2), (5), and (8) use as dependent variable an indicator for employment-to-employment (E2E) transitions, defined as job changes followed by an interruption of less than one month. Columns (3), (6), and (9) use as dependent variable an indicator for employment-to-unemployment (E2U) transitions, defined as job changes followed by an interruption of more than one month. Robust standard errors are reported in parentheses.

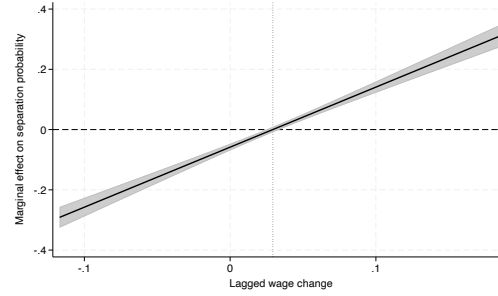
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Online Appendix Figure OA.3: Marginal Effect of Wage Changes on Subsequent Separations - E2E and E2U transitions

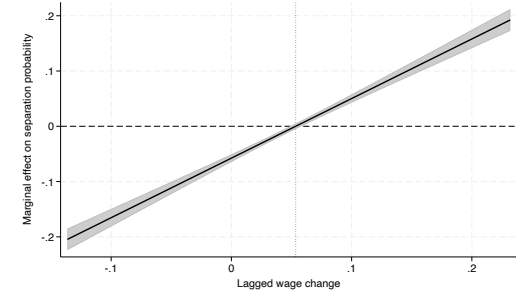
Employment to Employment Transitions



(a) Italy

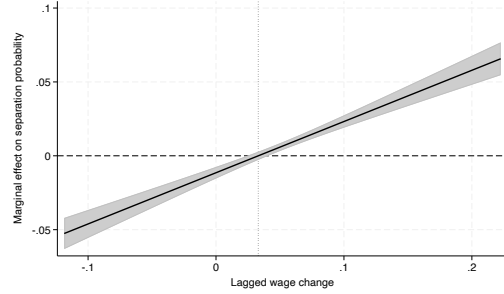


(b) Germany

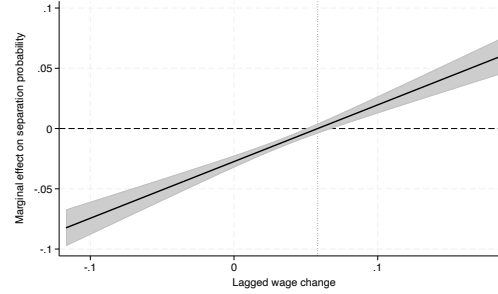


(c) Austria

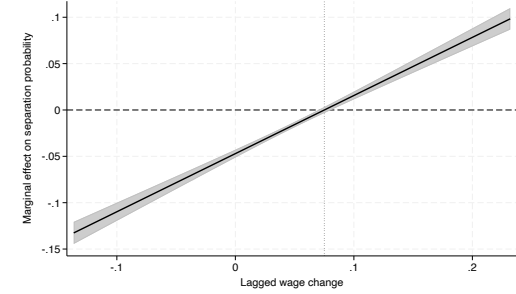
Employment to Unemployment Transitions



(d) Italy



(e) Germany

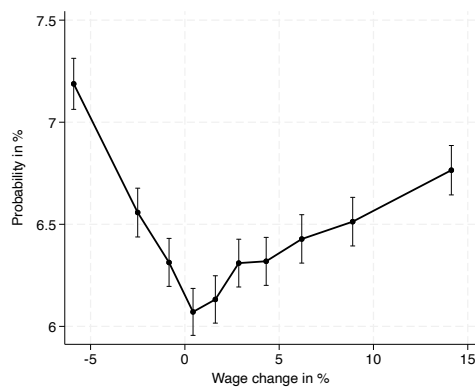


(f) Austria

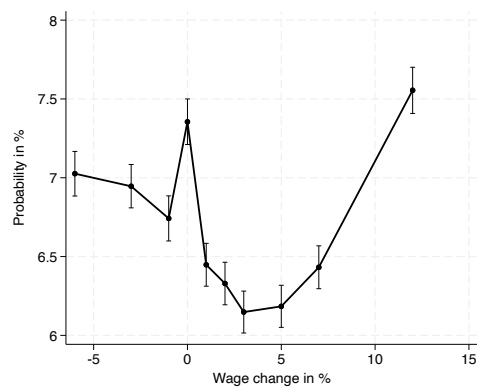
Notes: The figure plots the marginal effect of the lagged wage change on the probability of separation. For the quadratic specification, the marginal effect at wage change Δw is $\hat{\beta}_1 + 2\hat{\beta}_2\Delta w$. The solid line reports the estimated marginal effect and the shaded region shows the 95% pointwise confidence interval constructed from the joint variance-covariance matrix of the linear and quadratic coefficients. The horizontal dashed line marks a marginal effect of zero. The vertical dotted line marks the implied trough, $-\hat{\beta}_1/(2\hat{\beta}_2)$. Each panel is evaluated over the observed support of lagged wage changes in the corresponding regression sample.

Online Appendix Figure OA.4: Wage Changes and Subsequent Separations by Type of Separation - E2E and E2U transitions

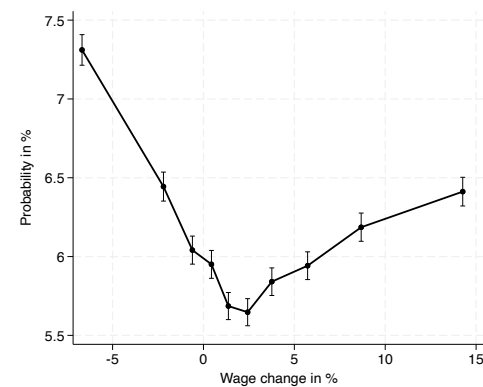
Employment to Employment Transitions



(a) Italy

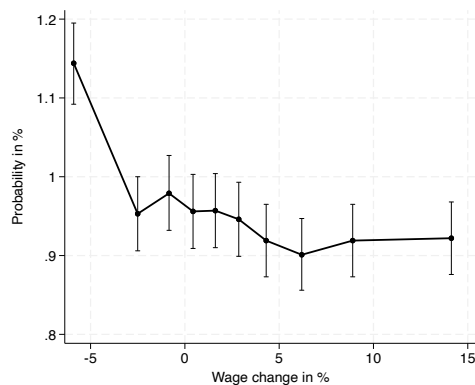


(b) Germany

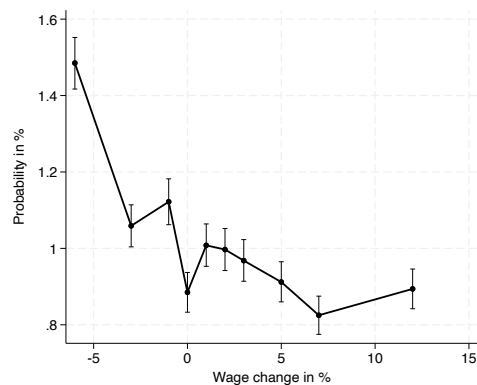


(c) Austria

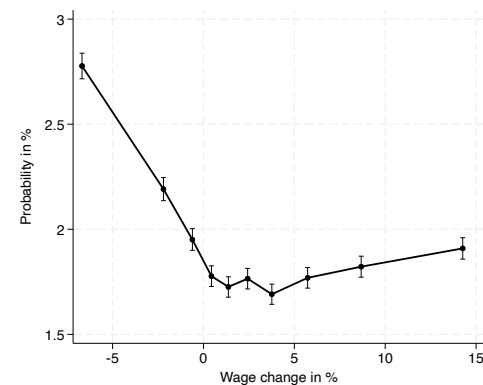
Employment to Unemployment Transitions



(d) Italy



(e) Germany



(f) Austria

Notes: The figure plots separation in year t against wage growth in year $t - 1$ by country, separately for employment to employment (E2E) and employment to unemployment (E2U) transitions. The solid line and markers show non-parametric within-bin separation rates (left axis). Wage changes are measured in log points. Error bars are computed from the corresponding linear probability model with wage-change-bin indicators. Each marker corresponds to one wage-change decile bin.

Online Appendix Table OA.14: Wage Changes and Subsequent Separations: Likely Layoffs and Quits in Germany

		1 month def.		3 month def.	
	Baseline	Layoff	Quit	Layoff	Quit
	(1)	(2)	(3)	(4)	(5)
ΔW_{t-1}	-0.092*** (0.0062)	-0.022*** (0.0022)	-0.070*** (0.0059)	-0.024*** (0.0023)	-0.068*** (0.0058)
ΔW_{t-1}^2	1.29*** (0.063)	0.17*** (0.021)	1.12*** (0.060)	0.19*** (0.022)	1.10*** (0.060)
W_{t-1}	-0.027*** (0.00087)	-0.019*** (0.00032)	-0.0072*** (0.00082)	-0.021*** (0.00033)	-0.0061*** (0.00082)
Obs.	1,229,924	1,229,924	1,229,924	1,229,924	1,229,924

Notes: The table reports coefficients from linear probability models of separation in year t on lagged wage changes and lagged log wages. Separations are divided in likely layoffs and quits, where a layoff is defined as receiving unemployment benefits within one month (three months) of separation. Quits are defined as the complement. Columns (1) uses the baseline separation indicator. Columns (2) and (4) use as dependent variable an indicator for likely layoff transitions. Columns (3) and (5) use as dependent variable an indicator for likely quit transitions. Columns (2) and (3) classify transitions using a one-month interruption threshold, while Columns (4) and (5) use a three-month interruption threshold. Robust standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Relative occupational wage position. Controlling for workers' relative wage position within occupation leaves the quadratic wage-growth coefficient essentially unchanged (Online Appendix Table OA.15). This matters because [Groes, Kircher and Manovskii \(2015\)](#) and [Perticara \(2004\)](#) document a U-shaped relationship between relative occupational wage position and occupational switching. Our result could therefore reflect occupational sorting rather than wage changes per se. To address this possibility, we control for relative occupational wage position, defined as $W_{t-1} - \mu_o(W_{t-1})$, as well as its square, where μ_o is the mean log wage in occupation o . For Germany we use three-digit occupations; for Italy, sector-by-qualification groups; for Austria, where occupation is unobserved, sector. Online Appendix Table OA.16 confirms that the occupational-wage-position U-shape is present in Germany and Italy. Online Appendix Table OA.15 shows that controlling for it does not affect the wage-change U-shape.

Search-intensity channel. Among movers, the probability of a wage gain upon transition declines in both tails of the prior wage-change distribution (Online Appendix Table OA.17). This is not what one would expect if endogenous search intensity were the main source of the U-shape. In models with endogenous on-the-job search, as in [Topa et al. \(2016\)](#), workers at low rungs of the job ladder may face both greater wage variability, because match quality and worker ability are still being learned, and higher offer arrival rates, because active search generates outside offers. If this mechanism were driving the right tail, workers experiencing

Online Appendix Table OA.15: Wage Changes and Subsequent Separations: Controlling for Occupational Wage Position

	Italy		Germany		Austria	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔW_{t-1}	-0.051*** (0.0052)	-0.051*** (0.0052)	-0.092*** (0.0062)	-0.10*** (0.0062)	-0.10*** (0.0040)	-0.11*** (0.0040)
ΔW_{t-1}^2	0.96*** (0.042)	0.96*** (0.042)	1.29*** (0.063)	1.28*** (0.063)	0.85*** (0.030)	0.88*** (0.030)
W_{t-1}	-0.061*** (0.00070)	-0.052*** (0.0012)	-0.027*** (0.00087)	0.020*** (0.0012)	-0.066*** (0.00055)	-0.11*** (0.0011)
$W_{t-1} - \mu(W_{t-1})^o$		-0.015*** (0.0015)		-0.078*** (0.0015)		0.060*** (0.0012)
$(W_{t-1} - \mu(W_{t-1})^o)^2$		0.013*** (0.0015)		0.048*** (0.0022)		-0.012*** (0.0011)
Obs.	1,645,579	1,645,579	1,229,924	1,229,924	2,758,316	2,758,316

Notes: The table reports coefficients from linear probability models of separation in year t on lagged wage changes and lagged log wages. Columns (1), (3), and (5) report the baseline specification. Columns (2), (4), and (6) additionally control for the worker's lagged relative wage position within occupation, entered linearly and quadratically. For Austria, where occupation is not observed, the corresponding control is based on the worker's position within the sectoral wage distribution. Robust standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

large wage changes should be disproportionately likely to move to higher-paying jobs. Instead, the share of upward-paying transitions is lowest in the tails. This evidence is difficult to reconcile with the simplest endogenous-search interpretation, although it does not rule out richer models in which search effort rises without systematically improving match quality.

Sensitivity. Two additional results speak to the stability of the finding. First, replacing lagged wage growth with contemporaneous wage growth, $\Delta W_t = W_t - W_{t-1}$, produces a more pronounced U-shape in both tails (Online Appendix Table OA.6 columns (2), (5) and (8)). We do not treat this specification as preferred, since contemporaneous wage growth is potentially more exposed to anticipatory wage adjustments. Second, the U-shape holds separately in East and West Germany, with somewhat larger magnitudes in the East (Online Appendix Table OA.12, columns (3)–(6)).²⁹ The West German estimates are quantitatively similar to those for Italy and Austria.

²⁹The IAB Establishment Panel includes East Germany only from 1996 onward; the 1993–1995 observations cover West Germany only (Fischer et al., 2014). The East–West comparison therefore pertains to 1996–2010.

Online Appendix Table OA.16: Wage Changes, Occupational Switching, and Separations

	Italy		Germany		Austria	
	Δ Occ. (1)	Sep. (2)	Δ Occ. (3)	Sep. (4)	Δ Sec. (5)	Sep. (6)
ΔW_{t-1}	0.015*** (0.0033)	-0.051*** (0.0052)	-0.056*** (0.0049)	-0.10*** (0.0062)	-0.048*** (0.0025)	-0.11*** (0.0040)
ΔW_{t-1}^2	0.68*** (0.029)	0.96*** (0.042)	0.94*** (0.050)	1.28*** (0.063)	0.33*** (0.018)	0.88*** (0.030)
W_{t-1}	-0.0063*** (0.00086)	-0.052*** (0.0012)	-0.0083*** (0.00085)	0.020*** (0.0012)	-0.032*** (0.00063)	-0.11*** (0.0011)
$W_{t-1} - \mu(W_{t-1})^o$	-0.0011 (0.0011)	-0.015*** (0.0015)	0.0019 (0.0011)	-0.078*** (0.0015)	0.012*** (0.00074)	0.060*** (0.0012)
$(W_{t-1} - \mu(W_{t-1})^o)^2$	0.080*** (0.0017)	0.013*** (0.0015)	0.051*** (0.0018)	0.048*** (0.0022)	-0.0036*** (0.00062)	-0.012*** (0.0011)
Obs.	1,645,579	1,645,579	1,229,924	1,229,924	2,758,316	2,758,316

Notes: The table reports coefficients from linear probability models relating lagged wage changes to occupational/sectoral switching and separation in year t . Columns (1), (3), and (5) use as dependent variable an indicator for an occupational/sectoral change; for Austria, the corresponding outcome is a sectoral change. Columns (2), (4), and (6) use as dependent variable an indicator for separation. All specifications additionally control for the worker's lagged relative wage position within occupation, entered linearly and quadratically. For Austria, the corresponding control is based on the worker's position within the sectoral wage distribution. Robust standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Online Appendix Table OA.17: Wage Gains upon Job-to-Job Mobility

	Italy	Germany	Austria
	(1)	(2)	(3)
1st decile	0.94* (0.024)	0.85*** (0.024)	0.92*** (0.017)
2nd decile	0.93** (0.025)	0.91** (0.026)	0.95** (0.018)
3rd decile	1.00 (.)	0.75*** (0.022)	0.94** (0.019)
4th decile	0.95 (0.026)	0.60*** (0.017)	0.95* (0.019)
5th decile	0.97 (0.026)	0.93* (0.028)	1.00 (.)
6th decile	0.93** (0.025)	1.00 (.)	0.92*** (0.019)
7th decile	0.86*** (0.023)	0.98 (0.029)	0.95** (0.019)
8th decile	0.83*** (0.022)	0.96 (0.029)	0.89*** (0.018)
9th decile	0.77*** (0.020)	0.94* (0.028)	0.84*** (0.017)
10th decile	0.71*** (0.019)	0.85*** (0.024)	0.76*** (0.015)
Obs.	122,264	98,957	223,511
Mean dep. var.	0.60	0.56	0.58

Notes: The table reports odds ratios from logit regressions of an indicator for positive wage growth upon job mobility on deciles of lagged wage changes. The sample is restricted to job switchers. The omitted category is the wage-change bin with the highest share of workers experiencing wage growth upon job mobility. Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

OA.1.3 Exact between–within decomposition

Let Y denote separation, X the lagged within-job wage change, and L the lagged wage. Write $C = (1, X, L)'$ and define

$$\pi = \arg \min_b E[(X^2 - C'b)^2], \quad Q = X^2 - C'\pi, \quad D = E[Q^2]. \quad (\text{OA.1})$$

Assume that the required fourth moments exist and $D > 0$.

Proposition OA.1.1 (Between–within decomposition). *For any discrete grouping variable G , the coefficient on X^2 in the population projection of Y on X^2 and C satisfies*

$$\beta_2 = B_G + W_G, \quad (\text{OA.2})$$

where

$$B_G = \frac{\text{Cov}(E[Q \mid G], E[Y \mid G])}{D}, \quad (\text{OA.3})$$

$$W_G = \frac{E[\text{Cov}(Q, Y \mid G)]}{D}. \quad (\text{OA.4})$$

Proof. The Frisch–Waugh–Lovell theorem gives $\beta_2 = E[QY]/E[Q^2]$. Because Q is a projection residual, $E[Q] = 0$, and hence $E[QY] = \text{Cov}(Q, Y)$. The law of total covariance gives

$$\text{Cov}(Q, Y) = \text{Cov}(E[Q \mid G], E[Y \mid G]) + E[\text{Cov}(Q, Y \mid G)].$$

Dividing by D proves the result. $\square$

The identity applies to tenure, experience, or multidimensional cells. Its interpretation depends on the grouping variable, and decompositions using different groupings are not additive. Moreover, W_G is a covariance contribution under one global residualization and denominator. It is not an average of coefficients estimated separately within groups.

The within component can also be partitioned over disjoint regions of the wage-change support without changing the residual or denominator. Let $\{\mathcal{S}_s\}_{s=1}^S$ partition the support of X and define

$$W_{G,s} = \frac{E[\mathbf{1}\{X \in \mathcal{S}_s\}(Q - E[Q \mid G])(Y - E[Y \mid G])]}{D}. \quad (\text{OA.5})$$

Because the support indicators sum to one,

$$\sum_s W_{G,s} = \frac{E[\text{Cov}(Q, Y \mid G)]}{D} = W_G. \quad (\text{OA.6})$$

This is an additive decomposition of the common coefficient, not a collection of regressions estimated on support-specific subsamples.

Consider the squared-only projection $Y = a + \tilde{\beta}_2 X^2 + e$. Then

$$\tilde{\beta}_2 = \frac{\text{Cov}(\mathbb{E}[X^2 | G], \mathbb{E}[Y | G]) + \mathbb{E}[\text{Cov}(X^2, Y | G)]}{\text{Var}(X^2)}. \quad (\text{OA.7})$$

Corollary OA.1.1.1 (Pure-composition benchmark). *Suppose $\mathbb{E}[X | G] = 0$ and $\text{Cov}(X^2, Y | G) = 0$ almost surely. Then*

$$\tilde{\beta}_2 = \frac{\text{Cov}(\text{Var}(X | G), \Pr(Y = 1 | G))}{\text{Var}(X^2)}. \quad (\text{OA.8})$$

Under these additional assumptions, tenure profiles that are both weakly decreasing generate a nonnegative squared-magnitude coefficient. Equation (OA.7) also shows why this is not an equivalence. A within-tenure covariance can reinforce or offset the between component, and curvature can arise without variation across tenure groups. Conditional independence, $X \perp\!\!\!\perp Y | G$, is sufficient for the covariance restriction but is stronger than required. The benchmark does not generally apply to the full model because news about worker ability and match productivity can affect both wages and mobility decisions within a tenure group.

OA.1.4 Decomposition Results for Germany and Austria

Table OA.18 reports the corresponding decomposition for Germany and Austria and compares the two Italian samples.

OA.1.5 Alternative Career-Stage Groupings

Table OA.19 repeats the exact decomposition using tenure, experience, and joint tenure-by-experience cells in each country. Within a country, the regression, sample, residual Q , and denominator are held fixed across rows; only the grouping variable G changes. Cells are weighted by their shares of observations.

The joint-cell decomposition preserves the cross-country contrast. Within-cell covariance accounts for more than three quarters of the controlled curvature in Italy and Germany. In Austria, 62.8 percent instead lies between joint tenure-by-experience cells. The different roles of career composition across countries therefore persist when career stage is allowed to vary flexibly with both tenure and experience. Again, we contrast the estimation sample results for Italy with those run on the empirical sample showing that they are quantitatively and qualitatively similar.

Online Appendix Table OA.18: Anatomy of the Wage-Change U-Shape across Countries

Statistic	Italy		Germany (3)	Austria (4)
	Estimation (1)	Empirical (2)		

<i>Panel A. Exact decomposition</i>				
Quadratic coefficient, $\widehat{\beta}_2$	95.618	95.609	129.013	85.238
Worker-clustered standard error	(4.237)	(4.179)	(6.403)	(3.027)
Between tenure, B_G	19.421	19.988	29.642	72.811
Within tenure, W_G	76.197	75.620	99.371	12.427
Negative wage changes, W_G^-	53.767	53.421	47.848 [†]	42.240
Zero wage changes, W_G^0	0.006	0.007	—	0.002
Positive wage changes, W_G^+	22.424	22.193	51.523	−29.815
<i>Panel B. U-shape moments</i>				
Between contribution ratio, $B_G/\widehat{\beta}_2$ (%)	20.31	20.91	22.98	85.42
Negative-within ratio, W_G^-/W_G (%)	70.56	70.64	48.15 [†]	339.91
Fitted slope at $X = 0.10$, $g_{0.1}$	13.982	13.980	16.594	6.578
Right-tail mass, $100 \Pr(X > 0.10)$ (%)	12.19	12.07	8.00	11.69
<i>Panel C. Sample</i>				
Observations	1,610,462	1,645,579	1,229,924	2,758,316

Notes: The dependent variable is an indicator for subsequent employer separation. The quadratic coefficient and its components are in percentage points per squared log point, and the fitted slope is in percentage points per log point. The support components use the common country-specific residual Q , denominator, and tenure-cell means; they are not coefficients from sign-specific regressions. The ratios in Panel B are signed contribution ratios and therefore need not lie between zero and one. Within each column, $B_G + W_G = \hat{\beta}_2$ and $W_G^- + W_G^0 + W_G^+ = W_G$, up to rounding.

Columns (2), (3) and (4) use the sample of Table 1, the baseline empirical-analysis sample. Column (1) uses the model-consistent estimation sample (Section 6.1).

[†] Due to data export restrictions of the FDZ, column (3) was produced with wage changes of exactly zero pooled into wage cuts, so W_G^- there reports the combined non-positive component and the identity is $W_G^- + W_G^+ = W_G$.

Online Appendix Table OA.19: Between–Within Decomposition across Alternative Career-Stage Cells

Grouping	Cells	B_G	W_G	Between share	Within share
<i>Panel A.1. Italy (estimation sample)</i>					
Tenure	18	19.421	76.196	20.31%	79.69%
Experience	17	15.359	80.259	16.06%	83.94%
Tenure $\times$ experience	169	22.194	73.423	23.21%	76.79%
<i>Panel A.2. Italy (empirical sample)</i>					
Tenure	18	19.988	75.620	20.91%	79.09%
Experience	17	16.466	79.143	17.22%	82.78%
Tenure $\times$ experience	169	23.045	72.563	24.10%	75.90%
<i>Panel B. Germany</i>					
Tenure	16	29.642	99.371	22.98%	77.02%
Experience	15	7.307	121.706	5.66%	94.34%
Tenure $\times$ experience	126	25.562	103.451	19.81%	80.19%
<i>Panel C. Austria</i>					
Tenure	16	72.811	12.427	85.42%	14.58%
Experience	15	37.667	47.572	44.19%	55.81%
Tenure $\times$ experience	135	53.500	31.738	62.77%	37.23%

Notes: Components are in percentage-point units and sum to $\widehat{\beta}_2$ up to rounding; shares divide each component by $\widehat{\beta}_2$. Cells are weighted by their shares of observations. Within a country, rows use the same sample, residual, and denominator. Because the groupings overlap, their decompositions are not additive.

Panel A.2., B and C use the sample of Table 1, the baseline empirical-analysis sample. Panel A.1. uses the model-consistent estimation sample (Section 6.1).

OA.2 Additional Details: Estimation

OA.2.1 Estimator, Sample, and Reporting Units

The estimator uses the 21 moments in Table OA.21. Let $\tilde{m}_j = d_j m_j$ express moment j in its reporting unit. We minimize

$$J(\Theta) = \sum_{j=1}^{21} [\tilde{m}_j^s(\Theta) - \tilde{m}_j^d]^2. \quad (\text{OA.9})$$

Each transformed discrepancy receives equal weight. Probabilities and component shares enter in percentage points, log wage levels are multiplied by 100, and wage-growth autocorrelation is expressed as 100 times the correlation. The sample retains at most nineteen employed observations per worker, matching the simulation horizon; annual non-employment is measured over interior years of each retained worker history.

Estimation uses Metropolis–Hastings proposals for global optimization followed by local refinement under common random numbers. The global stage runs 500 independent chains of length 100 from a dispersed start bank, with Gaussian proposals in the normalized parameter space tuned to moderate acceptance rates. The most promising candidates then seed six local refinement chains. All searches use the following configuration: 500,000 simulated workers, nineteen periods, and fixed common-random-number seeds. Finalists are ranked using the mean moment vector from three simulations of 500,000 workers.

OA.2.2 Estimation Sample Comparison

Online Appendix Table OA.20: Empirical-Analysis and Estimation Samples

	Empirical analysis	Estimation
<i>Panel A. Sample size</i>		
Observations	1,645,579	1,610,462
<i>Panel B. Wage changes and subsequent separation</i>		
Separation rate (percent)	7.43	7.51
Linear term, $\hat{\beta}_1$	-5.14	-5.14
Curvature, $\hat{\beta}_2$	95.61	95.62
Slope at +0.10	13.98	13.98
Fitted trough	0.0269	0.0269
Mass above +0.10 (percent)	12.07	12.19

Notes: The empirical-analysis column reports the sample used for the reduced-form evidence. The estimation column reports the model-comparable sample used to construct the structural targets. Panel B applies the same regression of subsequent separation on lagged wage growth, its square, and lagged log wages to both samples.

OA.2.3 Moment Definitions

Online Appendix Table OA.21: Definitions of the Estimation Moments

Moment	Definition	Unit
Quadratic coefficient, β_2	$100\hat{\beta}_2$ in $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_L L + \varepsilon$	Percentage-point coefficient
Between-tenure share, $100B/\beta_2$	$100B/\beta_2$, where $B = \beta_2 - W$ and $W = 100E[(Q - E[Q G])(Y - E[Y G])]/E[Q^2]$	Percent
Negative-change share, $100W^-/W$	$100W^-/W$, where $W^- = 100E[1\{X < 0\}(Q - E[Q G])(Y - E[Y G])]/E[Q^2]$	Percent
Right-hand slope, $g_{0.1}$	$g_{0.1} = 100(\hat{\beta}_1 + 2 \times 0.10\hat{\beta}_2)$	Percentage-point slope
Right-tail mass, $100\Pr(X > 0.10)$	$100\Pr(X > 0.10)$	Percent
Employment-to-non-employment rate	$100\Pr(E_{t+1} = 0 E_t = 1)$	Percent
Annual non-employment rate	$100\Pr(E_t = 0)$	Percent
Separation rate	$100\Pr(E_{t+1} = 0 \text{ or new job}_{t+1} = 1 E_t = 1)$	Percent
Mean lagged log wage	$100E[w_{t-1}]$	Log points $\times 100$
SD log wage, experience > 6	$100\text{SD}(w_{t-1} \text{Exp}_{t-1} > 6)$	Log percentage points
SD wage growth, young stayers	$100\text{SD}(\Delta w_{t-1} \text{two-period stayer}, \text{Exp}_{t-1} \leq 3)$	Log percentage points
SD wage growth, young movers	$100\text{SD}(\Delta w_{t-1} \text{job-to-job mover}, \text{Exp}_{t-1} \leq 3)$	Log percentage points
SD log wage, experience ≤ 3	$100\text{SD}(w_{t-1} \text{Exp}_{t-1} \leq 3)$	Log percentage points
Wage-growth autocorrelation, $100\rho_{\Delta w}$	$100\text{Corr}(\Delta w_t, \Delta w_{t-1} \text{two-period stayer})$	Correlation $\times 100$
Mean entry wage from non-employment	$100E[w_t E_t = 1, E_{t-1} = 0]$	Log points $\times 100$
SD wage growth, mid-career stayers	$100\text{SD}(\Delta w_{t-1} \text{two-period stayer}, 3 < \text{Exp}_{t-1} \leq 6)$	Log percentage points
Firm-rank change gap	$100\{E[R_d - R_o E2E, X < 0] - E[R_d - R_o E2E, X > 0]\}$	Percentile-rank points
Wage-variance slope by origin-firm rank	100 times the origin-rank coefficient for normalized squared residual wage growth, controlling for lagged wage, tenure FE, and experience FE	Coefficient $\times 100$
Absolute wage-growth experience slope	Slope from $100 \Delta w_{t-1} $ on standardized experience	Log percentage points per SD
Origin-firm top-quintile mass	$100\Pr(R_o > 0.8)$	Percent
Wage-firm-rank mismatch dispersion	$100E[(R_w(\text{Exp}) - R_o)^2]$, with wage rank computed within experience cells	Squared rank $\times 100$

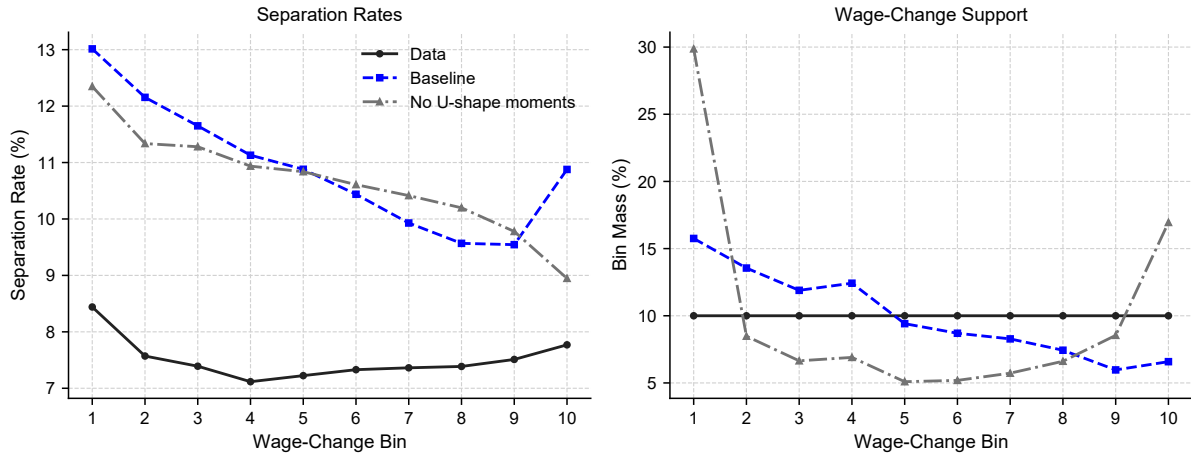
Moment	Definition	Unit
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Notes: The five U-shape moments and the four moments requiring an origin-firm rank use the U-shape regression sample, which trims wage changes at the second and ninety-eighth percentiles. The remaining internally estimated moments use the overall SMM panel, subject to the transition and experience restrictions stated in the definitions. Annual non-employment is the share of interior worker-years without an observed job; calendar gaps within retained worker spans count as non-employment. X is lagged wage growth, Y subsequent separation, L lagged log wage, G tenure, and Q the residual from projecting X^2 on $(1, X, L)$. E2E denotes a job-to-job transition. R_o and R_d are origin- and destination-firm percentile ranks, and $R_w(\text{Exp})$ is wage rank within an experience cell.

OA.2.4 Ten-Bin Levels and Support

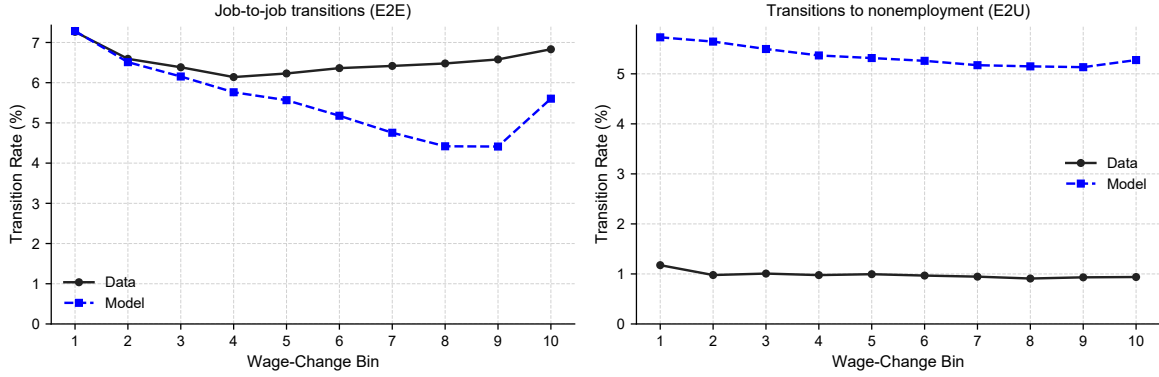
Figure OA.5 reports separation rates and probability mass in fixed empirical wage-change bins. The preferred estimate generates the right-tail rise, with a selected-sample separation rate of 11.22 percent compared with 7.51 percent in the data and a minimum farther to the right. The estimate without the U-shape moments places 19.08 percent of observations above 0.10 and generates a declining separation profile.

Online Appendix Figure OA.5: Ten-Bin Separation Rates and Wage-Change Support



Notes: Bins use fixed empirical wage-change decile cutoffs. The horizontal axis orders the common bins, whose widths in wage-change space vary. Model rates and masses average the three production simulations.

Online Appendix Figure OA.6: Transition Rates across the Wage-Change Distribution



Notes: The panels report unconditional E2E and E2U transition rates in the same empirical wage-change bins and selected sample as Figure 2. Model rates average three simulations of 500,000 workers. E2E and E2U are mutually exclusive and sum to total separation within each bin.

OA.2.5 Estimate without U-Shape Moments

Table OA.22 compares the preferred estimate with a reestimation based on the remaining sixteen moments. Omitting the U-shape moments lowers the point estimates of initial uncertainty and permanent ability dispersion, raises match-innovation volatility, reduces persistence, and lowers the employed offer probability. Initial uncertainty is estimated less precisely in the reestimation, while the confidence intervals for $\bar{\sigma}_\eta$, ρ , and λ_1 do not overlap across specifications.

Online Appendix Table OA.22: Parameter Estimates and Local Standard Errors with and without U-Shape Moments

Parameter	Preferred estimate		Without U-shape moments	
	Estimate	Local SE	Estimate	Local SE
Firm productivity SD, σ_μ	0.1327	0.00283	0.1247	0.00241
Match innovation SD, $\bar{\sigma}_\eta$	0.1225	0.00163	0.2897	0.00307
Mean ability, $\bar{a}$	4.9139	0.00623	4.9100	0.00644
Ability SD, σ_a	0.2014	0.00385	0.1843	0.00489
Initial uncertainty ratio, h	0.5090	0.01440	0.4824	0.05408
Nonemployed offer probability, λ_0	0.6640	0.00887	0.6694	0.00285
Employed offer probability, λ_1	0.1364	0.00187	0.1092	0.00148
Exogenous separation probability, δ	0.0541	0.00053	0.0560	0.00075
Bargaining weight, α	0.4665	0.01103	0.4991	0.00768
Match productivity persistence, ρ	0.9534	0.00476	0.8217	0.00602
Home-production flow, z	4.0341	0.02394	3.8976	0.01651
Initial uncertainty SD, $\sigma_{a,0}$	0.1025	0.00413	0.0889	0.01224

Notes: Both specifications use the eleven-parameter baseline model. The estimate without U-shape moments minimizes loss over the sixteen remaining moments. Local standard errors combine empirical worker-cluster sampling and finite-simulation uncertainty. They are specification-specific and do not constitute a direct test of equality across estimates. Initial uncertainty SD is the derived quantity $\sigma_{a,0} = h\sigma_a$.

Online Appendix Table OA.23: Sensitivity to the Weight on the U-Shape Moments

	No U	Quarter	Half	Preferred	Double
Relative U-shape weight, τ	0.00	0.25	0.50	1.00	2.00
Persistence, ρ	0.822	0.953	0.942	0.953	0.949
Innovation SD, $\bar{\sigma}_\eta$	0.290	0.123	0.136	0.123	0.112
Match-shock half-life (years)	3.5	14.5	11.6	14.5	13.2

Notes: The reweighted objective is $J_\tau = J_{-U} + \tau J_U$, where J_{-U} is loss over the other sixteen moments and J_U is loss over the five U-shape moments. Each column reports a separate local reestimation, except that the no-U and preferred columns reproduce the corresponding estimates in Table OA.22. All specifications set $\gamma = \kappa = 0$.

Online Appendix Table OA.24: Sensitivity of Economic Conclusions to the Weight on U-Shape Moments

	No U	Quarter	Half	Preferred	Double
Lifecycle decline (percent)	17.81	26.50	18.23	26.50	25.88
Learning contribution (p.p.)	3.03	22.78	17.67	22.78	20.04
Share of total decline (percent)	17.0	86.0	96.9	86.0	77.4
Negotiation-history contribution (p.p.)	16.74	6.18	1.91	6.18	7.67
Share of total decline (percent)	94.0	23.3	10.5	23.3	29.6
Other lifecycle forces (p.p.)	-1.96	-2.46	-1.35	-2.46	-1.83
Offer-decision disagreement (percent)	4.44	6.92	7.05	6.92	6.51
Aggregate output loss (percent)	0.073	0.229	0.191	0.229	0.179

Notes: Entries average three simulations of 500,000 workers using common worker-period shocks. Learning and negotiation-history contributions are two-factor Shapley values for the decline in stayer wage-change dispersion from 0–3 to 10 or more years of experience; together with other lifecycle forces, they sum to the total decline. Because other lifecycle forces can offset that decline, the learning and negotiation shares need not sum to 100 percent. Offer-decision disagreement is the share of on-the-job offers for which the learning and corresponding full-information rules select different actions. The aggregate output loss compares each learning economy with its separately solved full-information economy under the same parameter vector. Table OA.32 reports delta-method uncertainty for the no-U and preferred endpoints; the intermediate columns are weighting-sensitivity point comparisons.

OA.2.6 Fit under Reestimated Restrictions

Table OA.25 reports fit by moment block. Near-observed ability fixes the initial posterior ability standard deviation at $\sigma_{a,0} = 10^{-4}$. Setting $\rho = 0$ or $\alpha = 1$ produces much larger deteriorations, consistent with the U-shape outcomes reported in the main text.

Online Appendix Table OA.25: Fit by Moment Block under Exclusion and Model Restrictions

Specification	Other-moment loss	U-shape loss	Total loss
Preferred estimate	433.45	106.23	539.68
Without U-shape moments	251.95	106304.68	106556.63
Near-observed ability	443.53	579.72	1023.25
$\rho = 0$	1160.73	37262.19	38422.92
$\alpha = 1$	1676.39	101693.79	103370.18

Notes: The estimate without the U-shape moments minimizes loss over the sixteen remaining moments; the restricted specifications use all 21 moments. Near-observed ability fixes $\sigma_{a,0} = 10^{-4}$. All specifications set $\gamma = \kappa = 0$.

OA.2.7 Empirical Moment Uncertainty

Table OA.26 reports marginal standard errors for the twenty-one estimation targets, based on the 200 worker-cluster bootstrap draws that also deliver the empirical moment covariance

used in parameter inference (Section OA.2.8).

Online Appendix Table OA.26: Empirical Moment Sampling Uncertainty

Moment	Data	Bootstrap SE
Curvature, β_2	95.62	4.2828
Between-tenure share, $100B/\beta_2$	20.31	0.9789
Negative within share, $100W^-/W$	70.56	3.2852
Slope at +0.10	13.98	0.5889
Mass above +0.10 (percent)	12.19	0.0270
Employment to non-employment	1.86	0.0075
Annual non-employment	5.99	0.0261
Overall separation	14.23	0.0214
Mean lagged log wage	480.52	0.0429
SD log wage, high experience	30.84	0.0685
SD wage growth, young stayers	12.11	0.0304
SD wage growth, young movers	18.28	0.0566
SD log wage, young workers	22.63	0.0392
Wage-growth autocorrelation	-17.21	0.1106
Mean entry log wage from non-employment	469.57	0.1204
SD wage growth, mid-career stayers	9.51	0.0312
Firm-rank change gap	-2.55	0.2190
Wage-variance slope by origin rank	6.43	0.6492
Absolute wage-growth experience slope	-0.94	0.0055
Origin top-quintile mass	30.01	0.1025
Wage-firm-rank mismatch dispersion	7.49	0.0227

Notes: Standard errors are sample standard deviations across 200 worker-cluster bootstrap estimates, in the same reporting units as the targets. Firm effects, firm ranks, and wage-change cutoffs are held fixed.

OA.2.8 Local Parameter Inference

We estimate the empirical moment covariance, S_d , from 200 worker-cluster bootstrap draws of complete worker histories, holding firm effects, firm ranks, and wage-change cutoffs fixed. Simulation covariance, S_s , is estimated from 24 independent baseline simulations of 500,000 workers.

For each specification, let q denote normalized parameter coordinates, $G = \partial \tilde{m}^s / \partial q'$ its $K \times 11$ moment Jacobian, and $H = \partial r(q) / \partial q'$ the transformation to reported parameters. Here $K = 21$ for the preferred estimate and $K = 16$ for the estimate without U-shape moments. With equal weights in reporting units, $A = (G'G)^{-1}G'$, and the local covariance is

$$\hat{V}_{\text{combined}} = HA(S_d + S_s/3)A'H'. \quad (\text{OA.10})$$

The divisor of three reflects the three simulations averaged by the estimator. Volatility

parameters are transformed from log standard deviations to levels, and the transformation for $\sigma_{a,0} = h\sigma_a$ retains covariance between h and σ_a . The intervals are conditional on the estimated firm effects, firm ranks, and wage-change cutoffs.

OA.2.9 Local Moment–Parameter Sensitivity

For a row set $\mathcal{A}$, let $G_{\mathcal{A},j}$ denote parameter j 's Jacobian column and $G_{\mathcal{A},-j}$ the remaining ten columns. Define

$$c_j(\mathcal{A}) = \min_{b \in \mathbb{R}^{10}} \|G_{\mathcal{A},j} - G_{\mathcal{A},-j}b\|^2, \quad L_j = 100 \left[1 - \frac{c_j(-U)}{c_j(21)} \right], \quad (\text{OA.11})$$

where 21 denotes all moments and $-U$ the remaining sixteen. Thus L_j is the percentage reduction in profiled local loss curvature when the U-shape moments are omitted.

Online Appendix Table OA.27: Contribution of the U-Shape Moments to Profiled Local Loss Curvature

Parameter direction	Reduction (%)	Step-size range
Firm productivity SD, σ_μ	65.0	64.3–68.9
Match innovation SD, $\bar{\sigma}_\eta$	76.2	73.7–81.2
Mean ability, $\bar{a}$	94.9	94.1–96.7
Ability SD, σ_a	90.6	89.9–91.9
Initial uncertainty ratio, h	38.7	31.4–40.6
Nonemployed offer probability, λ_0	13.5	13.3–14.6
Employed offer probability, λ_1	41.5	36.2–42.2
Exogenous separation probability, δ	38.1	38.1–46.3
Bargaining weight, α	84.2	82.2–84.8
Match productivity persistence, ρ	75.9	69.2–78.1
Home-production flow, z	62.6	59.4–62.6

Notes: Entries report L_j from equation (OA.11), in percent. Ranges span the eight retained step-size combinations.

The U-shape moments contribute most strongly to local discipline on mean ability, ability dispersion, bargaining power, match-innovation volatility, and persistence. This pattern accords with the nonlinear reestimation, which selects a substantially different stochastic process when these moments are omitted.

OA.2.10 Curvature under Fixed-Parameter Restrictions

The fixed-parameter exercises solve each restricted economy while holding every remaining primitive at the preferred estimate. Table OA.28 decomposes the resulting curvature. Near-observed ability removes the positive between-tenure contribution, $\rho = 0$ produces an inverted U-shape, and $\alpha = 1$ makes the right-hand within-tenure component negative.

Online Appendix Table OA.28: Curvature Components with Other Primitives Held Fixed

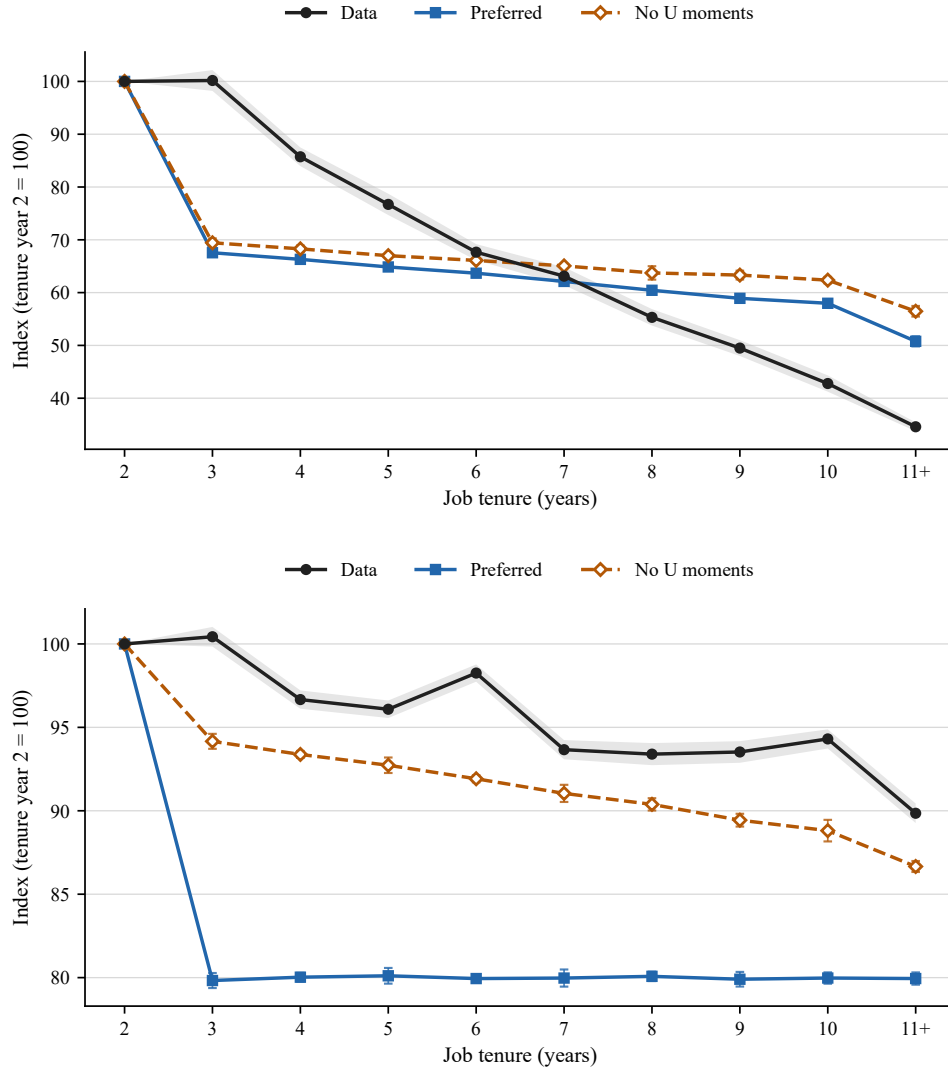
Restriction	β_2	B	W	W^-	W^0	W^+
Preferred estimate	96.56	20.36	76.20	51.95	-0.00	24.25
Near-observed ability	118.12	-2.51	120.63	73.67	-0.00	46.96
$\rho = 0$	-62.56	-24.32	-38.24	32.87	-0.00	-71.11
$\alpha = 1$	21.54	17.05	4.49	40.99	-0.00	-36.49

Notes: $B + W = \beta_2$ and $W^- + W^0 + W^+ = W$, up to rounding. Near-observed ability fixes $\sigma_{a,0} = 10^{-4}$. The small zero-change term is retained in the accounting identity.

OA.2.11 Untargeted Tenure Profiles

Both estimates reproduce the overall decline in separation with tenure but place too much of it between tenure years 2 and 3 and remain above the data at long tenure. The preferred estimate likewise front-loads the decline in wage-change dispersion, while the estimate without the U-shape moments follows its gradual empirical profile more closely.

Online Appendix Figure OA.7: Untargeted Tenure Profiles



Notes: The upper panel reports separation and the lower panel the standard deviation of wage changes. Each series is indexed to 100 in tenure year 2, the first year in which data and simulated wage changes share common support. The corresponding tenure-2 levels are 10.42, 16.54, and 15.43 percent for separation and 6.11, 7.29, and 10.91 percent for wage-change dispersion in the data, preferred estimate, and estimate without U-shape moments. Gray bands are worker-clustered 95 percent jackknife intervals; model error bars are 95 percent Monte Carlo intervals across the three production simulations.

OA.3 Firm-Dependent Learning and Worker–Firm Complementarity

This appendix allows signal precision and the return to worker ability to vary with firm productivity. Both specifications nest the baseline model.

OA.3.1 Extended Model

Let $Z_\mu = \mu/\sigma_\mu$, so that $Z_\mu \sim \mathcal{N}(0, 1)$ in the firm population. Firm-dependent match-innovation volatility is

$$\sigma_\eta(\mu) = \bar{\sigma}_\eta \exp(\gamma Z_\mu - \gamma^2), \quad E_\mu[\sigma_\eta(\mu)^2] = \bar{\sigma}_\eta^2. \quad (\text{OA.12})$$

Positive γ makes output less informative about worker ability at more productive firms. Worker–firm complementarity is introduced through

$$c(\mu; \kappa) = \exp(\kappa Z_\mu - \kappa^2), \quad E_\mu[c(\mu; \kappa)^2] = 1, \quad (\text{OA.13})$$

so log productivity contains $\bar{a} + c(\mu; \kappa)(a_i - \bar{a})$. Positive κ raises the return to worker ability at more productive firms. The normalizations hold the relevant population second moments fixed, and the conditional updating problem remains linear Gaussian. The baseline corresponds to $\gamma = \kappa = 0$.

OA.3.2 Matched Fit Comparison

We reestimate all thirteen parameters jointly on the same twenty-one moments and reporting weights as the baseline. The resulting estimates are $\hat{\gamma} = 0.003026$ and $\hat{\kappa} = 0.002097$. Between firms at the tenth and ninetieth percentiles of productivity, these values imply differences of 0.78 percent in match-innovation volatility and 0.54 percent in the loading on worker ability.

Online Appendix Table OA.29: Matched Fit of the Firm-Dependent Specifications

Specification	Other-moment loss	U-shape loss	Total loss
Extended model	430.6383	107.8584	538.4967
Baseline projection	433.9402	105.9788	539.9189

Notes: The first row reports the extended model. The second sets $\gamma = \kappa = 0$ while holding the remaining parameters at their extension estimates. Both rows use the same three simulations, moments, and reporting weights.

Setting both parameters to zero raises total loss by 0.264 percent and slightly reduces loss from the five U-shape moments. The specifications also yield nearly identical curvature, decompositions, right-hand slopes, and right-tail mass. Thus firm-dependent signal precision

and worker–firm complementarity contribute little to fit, supporting the constant-precision, additive baseline.

OA.3.3 Allocative Implications

We also compare each specification with its own full-information economy. This exercise holds the first eleven parameters fixed at the extension estimate and uses the same worker-period shocks across specifications.

Online Appendix Table OA.30: Learning Relative to Full Information in the Extended Model

Specification	Output loss (%)	Employment gap (pp)	Erroneous moves (%)	Missed moves (%)
Free γ and κ	0.233	0.013	3.34	3.61
Matched $\gamma = \kappa = 0$	0.233	0.012	3.34	3.61

Notes: The first row reports the extension estimate. The second sets $\gamma = \kappa = 0$ while holding the remaining parameters fixed. Each learning economy is compared with its separately solved full-information economy using the same three simulations of 500,000 workers. Output loss is full-information minus learning aggregate level output, as a percentage of full-information output. Erroneous and missed moves are percentages of on-the-job offer decisions.

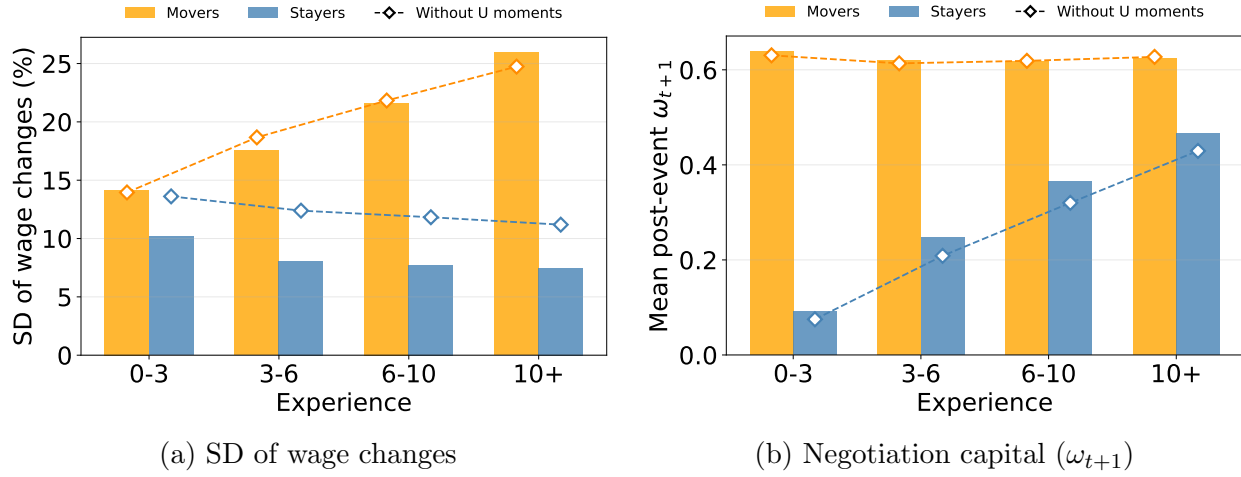
The extension and its matched zero projection both imply an aggregate output loss of 0.233 percent. Their losses differ by only 0.00010 percentage points, and their employment and offer-decision outcomes are also nearly identical. The aggregate allocative effect therefore operates through learning with persistent match productivity, the channel retained in the additive baseline.

OA.4 Additional Application Results

OA.4.1 Wage Changes of Movers and Stayers

Figure OA.8 separates wage changes within continuing jobs from those associated with job-to-job moves. Both estimates produce similar rising wage-change dispersion among movers, while stayer dispersion declines more sharply under the preferred estimate.

Online Appendix Figure OA.8: Wage-Change Volatility and Negotiation Capital: Movers versus Stayers



Notes: The figure is based on three simulated panels of 500,000 workers. Filled bars report the preferred estimate and hollow diamonds the no-U reestimation; orange denotes movers and blue denotes stayers. Panel (a) reports $SD(\Delta w_{t+1})$ for stayers and job-to-job movers by experience. Panel (b) reports mean negotiation capital ω_{t+1} after a potential move, conditional on employment at the next decision node.

OA.4.2 Employment-Flow Accounting

Table OA.31 decomposes the employment gap between each learning economy and its full-information benchmark. Learning reduces endogenous separations at the preferred estimate and raises them without the U-shape moments; entry from non-employment offsets most of either difference.

Online Appendix Table OA.31: Employment-Flow Accounting Relative to Full Information

	With U moments	Without U moments
Employment gap (p.p.)	0.013	-0.014
Entry-flow contribution (p.p.)	-0.107	0.090
Exogenous-exit contribution (p.p.)	-0.011	0.010
Endogenous-exit contribution (p.p.)	0.131	-0.115
U-E hazard, learning (%)	66.399	66.974
U-E hazard, FI (%)	66.382	66.972
Endogenous E2U hazard, learning (%)	0.482	0.767
Endogenous E2U hazard, FI (%)	0.492	0.754

Notes: Each column compares a learning economy with its own full-information benchmark, holding parameters and worker-period shocks fixed. The employment gap and flow contributions are in percentage points. Flow differences are weighted by the subsequent periods over which they affect employment, so the three contributions sum to the average employment gap. Entries average three simulations of 500,000 workers.

OA.4.3 Uncertainty in Application Outcomes

For each outcome $g(q)$ under each estimate, we apply the delta method using that specification's combined empirical and simulation parameter covariance. Central differences are evaluated under common worker-period shocks, and every endpoint averages three simulations of 500,000 workers.

Online Appendix Table OA.32: Delta-Method Uncertainty under Both Estimates

Outcome	With U moments		Without U moments	
	Estimate	95% CI	Estimate	95% CI
Learning contribution (p.p.)	22.78	[20.42, 25.14]	3.03	[1.35, 4.72]
Aggregate output loss (%)	0.229	[0.214, 0.245]	0.073	[0.035, 0.110]
Employment gap (p.p.)	0.013	[-0.005, 0.032]	-0.014	[-0.028, -0.001]
Tail decision-error change (p.p.)	0.528	[0.325, 0.730]	-0.342	[-0.390, -0.294]
Tail current-output-loss change (p.p.)	0.045	[0.029, 0.060]	-0.031	[-0.041, -0.020]

Notes: Combined standard errors propagate empirical sampling and finite-simulation uncertainty through local numerical derivatives. Each 95% confidence interval equals the estimate plus or minus 1.96 combined standard errors within the indicated specification. The intervals are specification-specific and are not tests of differences across specifications. The learning contribution is the Shapley contribution to the career decline in stayer wage-change dispersion. Employment and output gaps compare learning with full information; tail changes are differences from the ninth to the tenth wage-change decile.